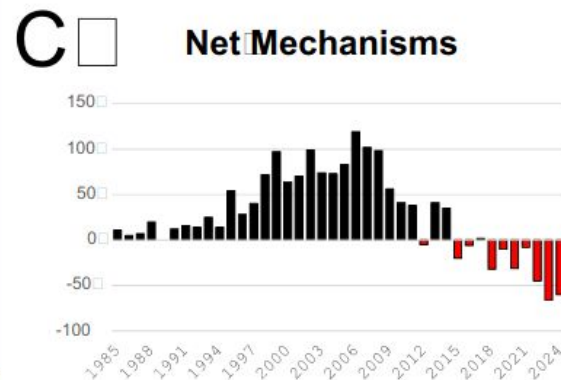
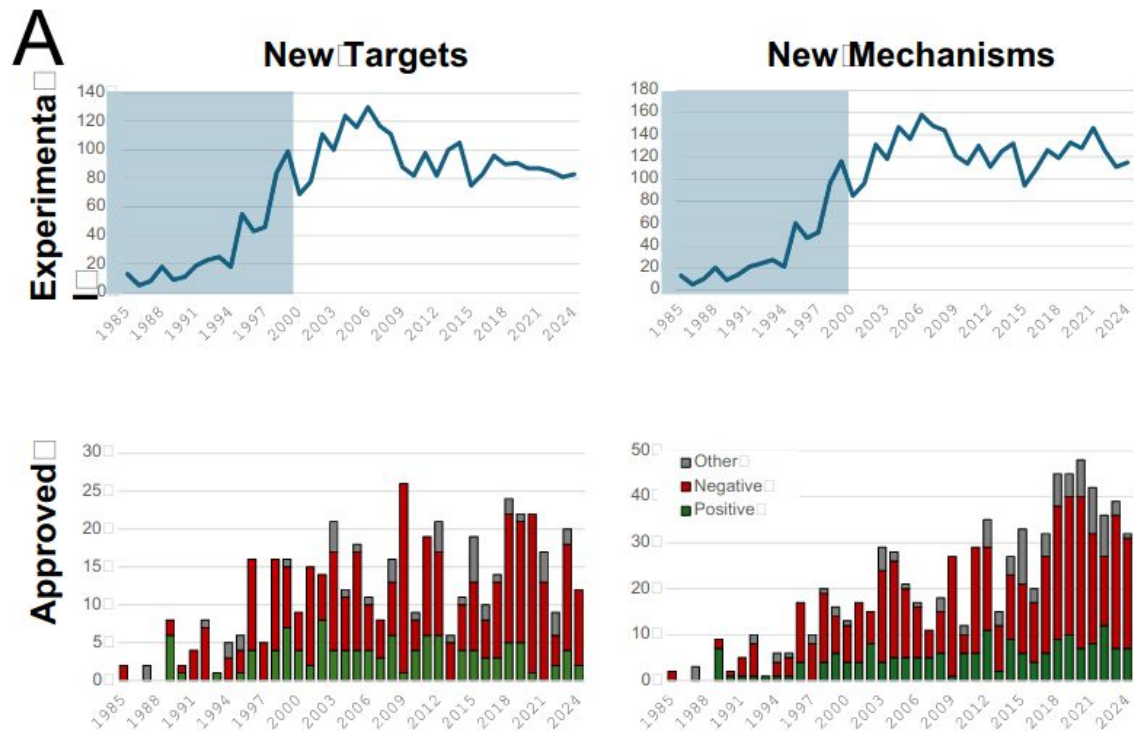




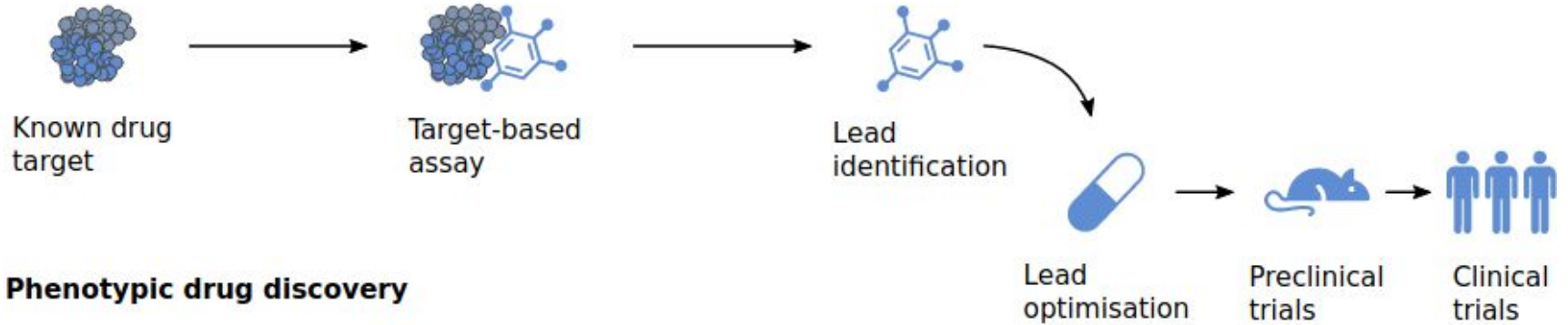
Kinch, Michael S., Zachary Kraft, and Tyler Schwartz. "Target & Mechanism Selection in Pharmaceutical Innovation: A Source for Concern or Optimism?" *Drug Discovery Today* 30, no. 5 (2025): 104361. <https://doi.org/10.1016/j.drudis.2025.104361>.

What can we do if there are no good targets

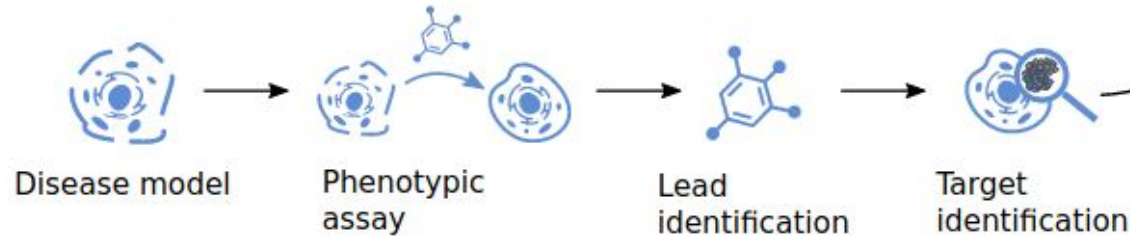
Mathematical and Computational Biology in Drug Discovery Module II

Dr. Jitao David Zhang
March-April 2026

Target-based drug discovery

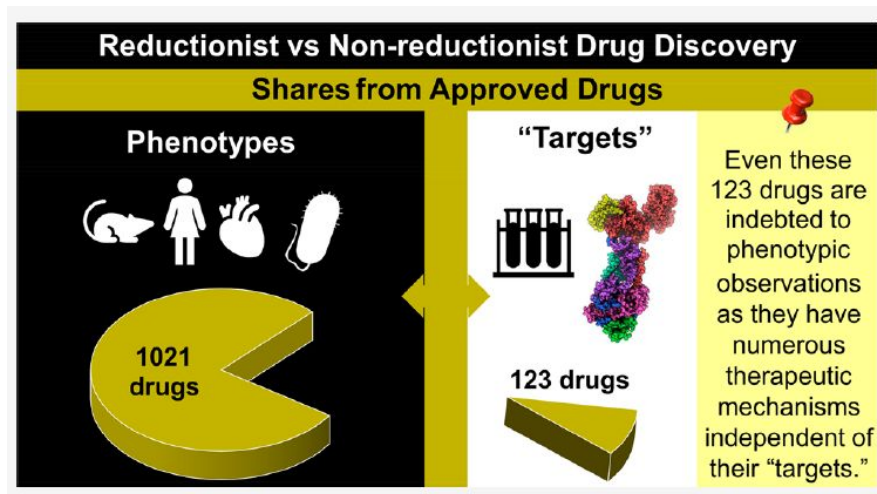


Phenotypic drug discovery

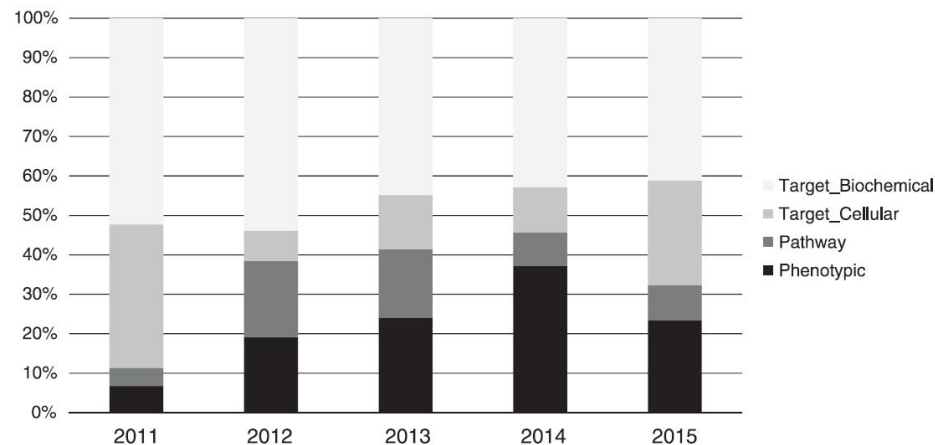


Krentzel, Daniel, Spencer L. Shorte, and Christophe Zimmer. "Deep Learning in Image-Based Phenotypic Drug Discovery." Trends in Cell Biology 33, no. 7 (2023): 538–54. <https://doi.org/10.1016/j.tcb.2022.11.011>.

Is target-based drug discovery the only way?



Sadri, Arash. 2023. "[Is Target-Based Drug Discovery Efficient? Discovery and 'Off-Target' Mechanisms of All Drugs.](#)" *Journal of Medicinal Chemistry* 66 (18): 12651–77.

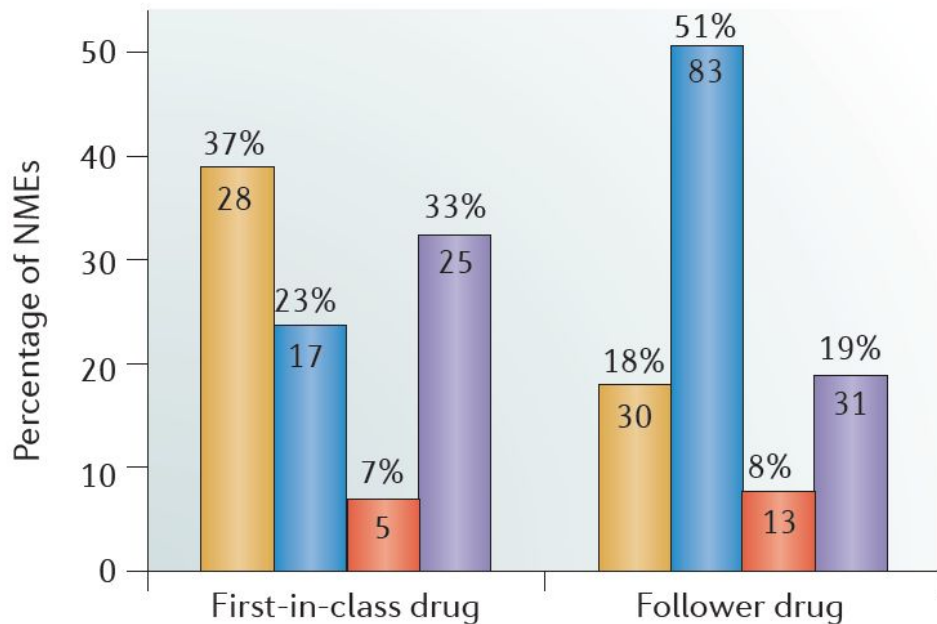


Haasen, Dorothea, Ulrich Schopfer, Christophe Antczak, Chantale Guy, Florian Fuchs, and Paul Selzer. 2017. "[How Phenotypic Screening Influenced Drug Discovery: Lessons from Five Years of Practice.](#)" *ASSAY and Drug Development Technologies* 15 (6): 239–46.

Five strategies when no good target is found

1. Phenotypic drug discovery
2. Natural products
3. Biologics
4. Interaction-based (multispecific) drug discovery
5. Drug repurposing or combination studies

Connect the lines!



- Phenotypic screening
- Modified natural products
- Biologics
- Target-based screening



Phenotypic screenings by agent and readout

Agent

High-throughput screening
libraries ($\geq 10^6$ molecules)

Genetic libraries ($\sim 10^4$)

Natural products and chemo-
genomic libraries ($\sim 10^3$)

Custom libraries ($\sim 10^0 - 10^2$)

**Boundary of
feasibility**

Reporters

Gene
expression

Cellular
morphology

Organ/tissue
phenotype

Organism
phenotype

Readout

Genetic screening

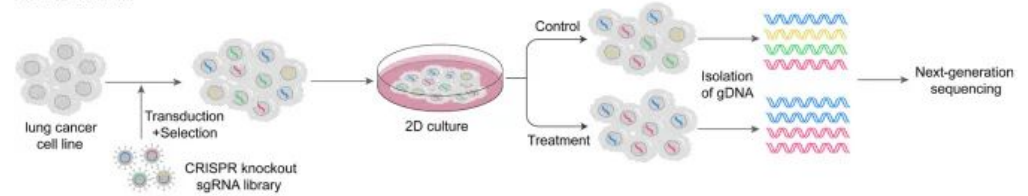
CRISPR (clustered regularly interspaced short palindromic repeats) screening is commonly used to identify potential targets in cellular or animal disease models.

Illustrated example: applications in lung cancer.

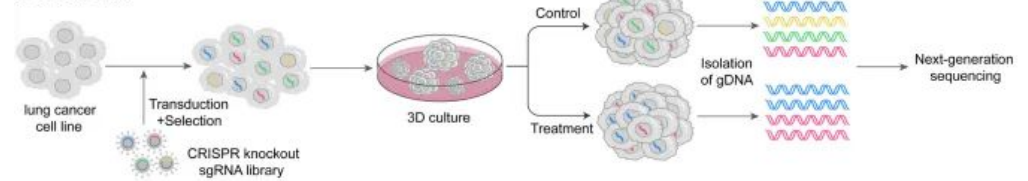
Shen, Wanying, Fangli Hu, Pan Lei, and Yijun Tang. "Applications of CRISPR Screening to Lung Cancer Treatment." *Frontiers in Cell and Developmental Biology* 11 (December 2023).
<https://doi.org/10.3389/fcell.2023.1295555>.

A In vitro CRISPR Screen

a 2D in vitro

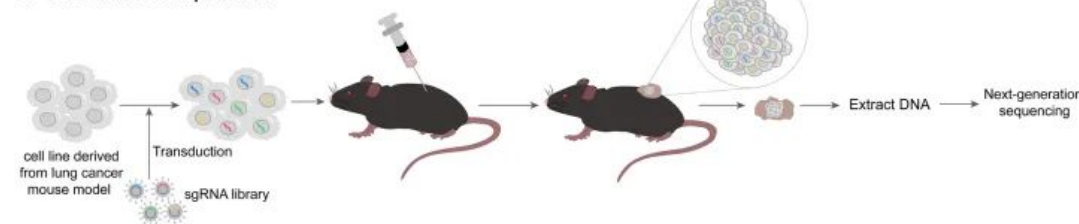


b 3D in vitro

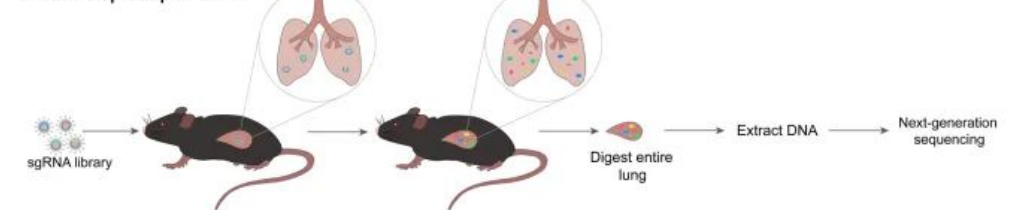


B In vivo CRISPR Screen

a Subcutaneous implantation



b Orthotopic implantation



Alternative: chemical library screening

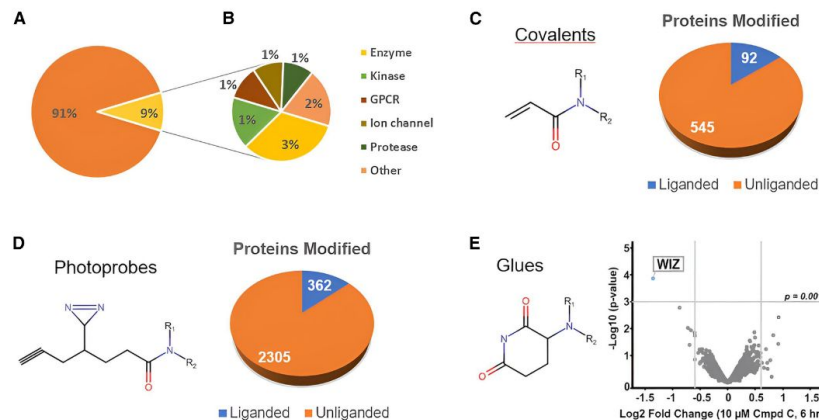
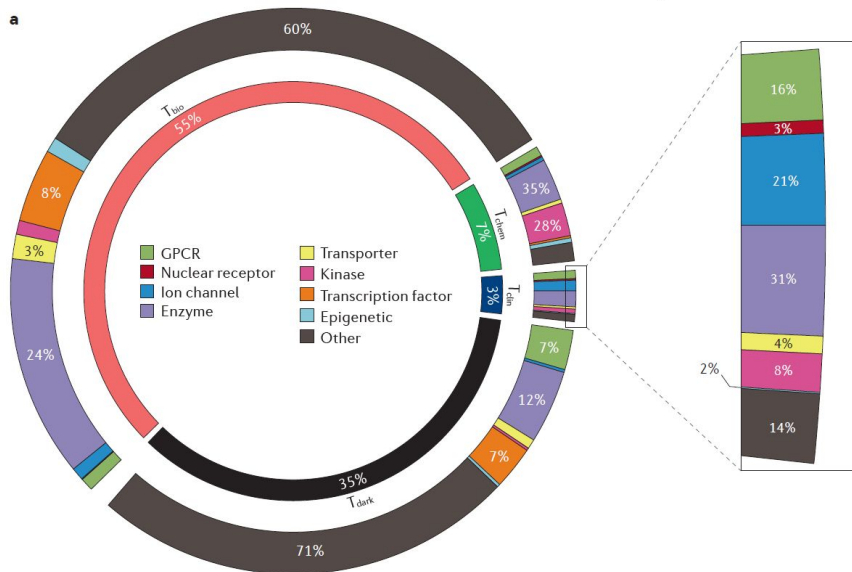


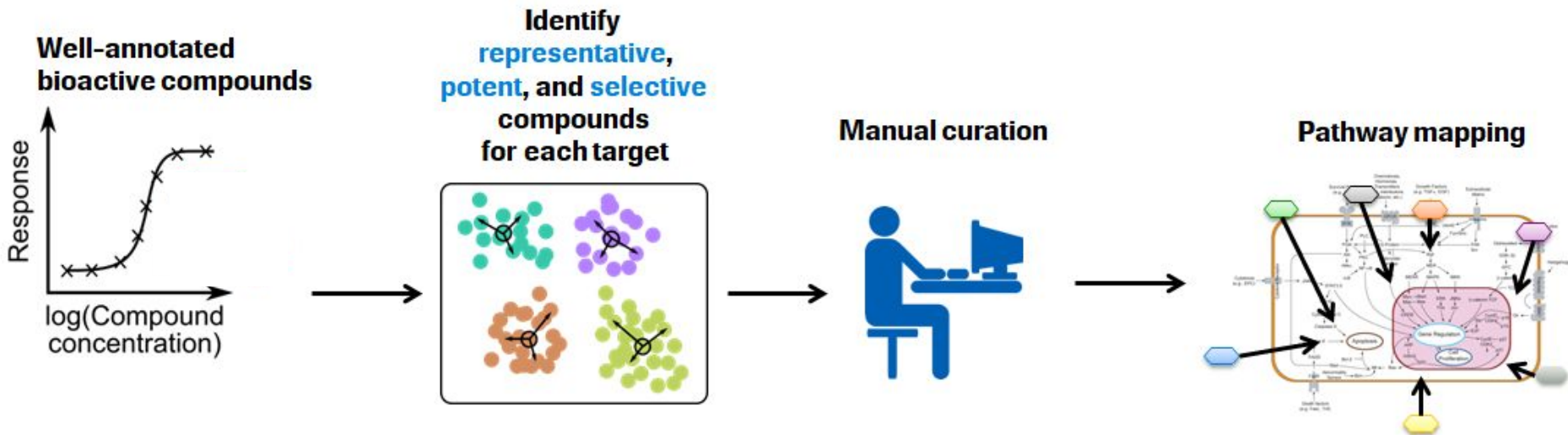
Figure 1. Genome coverage of compound libraries

(A) Coverage of the most extensive chemogenomics library published to date as a fraction of the human genome and (B) target classes covered in the library; adapted from Canham et al.⁸ Expanding biological space through new types of compounds including (C) covalent (liganded status based on DrugBank), (D) photoreactive (liganded status based on T clin + T chem vs. T bio + T dark),⁹ and (E) molecular glues; adapted and reproduced with permission from Backus et al.,¹⁰ Offensperger et al.,¹¹ and Ting.¹²

Left: Oprea, Tudor I., Cristian G. Bologa, Søren Brunak, Allen Campbell, Gregory N. Gan, Anna Gaulton, Shawn M. Gomez, et al. 2018. "Unexplored Therapeutic Opportunities in the Human Genome." *Nature Reviews Drug Discovery* 17 (February): 317–32. <https://doi.org/10.1038/nrd.2018.14>; Right: Vincent, Fabien, and Davide Gianni. "The Limitations of Small Molecule and Genetic Screening in Phenotypic Drug Discovery." *Cell Chemical Biology* (2025). <https://doi.org/10.1016/j.chembiol.2025.10.008>.

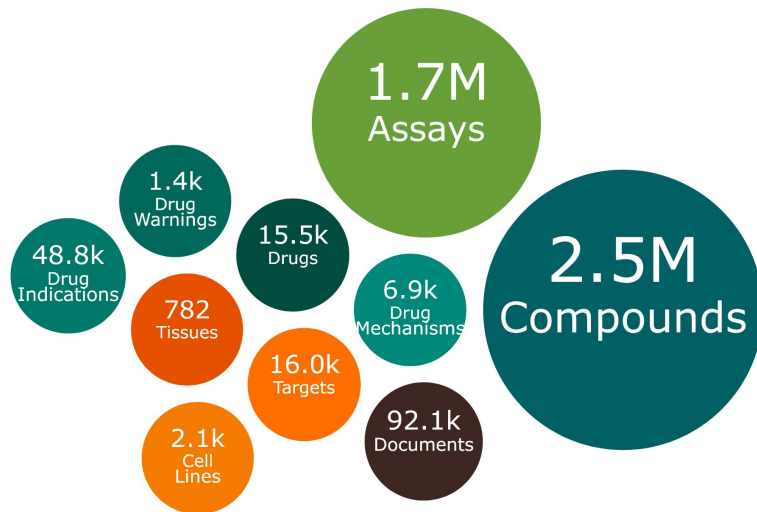
The Small-molecule Pathway Research Kit (SPARK)

Now known as the Pathway Annotated Chemical Ensemble (PACE) library



The ChEMBL database

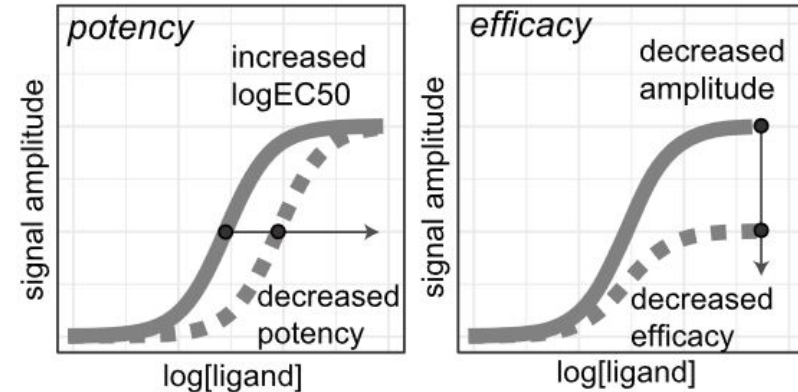
- An example of query: [aspirin](#).
- Systematic and programmatic accession via [ChEMBL API](#) ([source code](#)).
- We can use **dose-response data** to annotate the *triplets* of compound, assay activity, and targets.



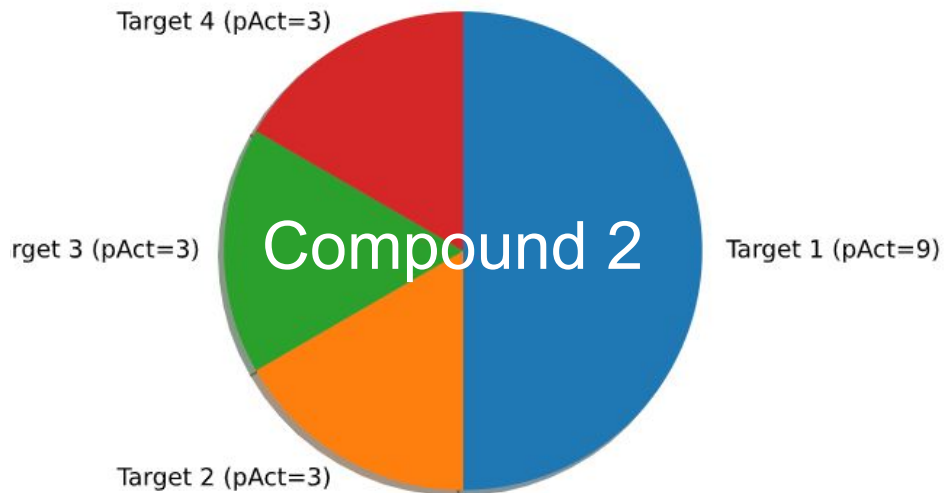
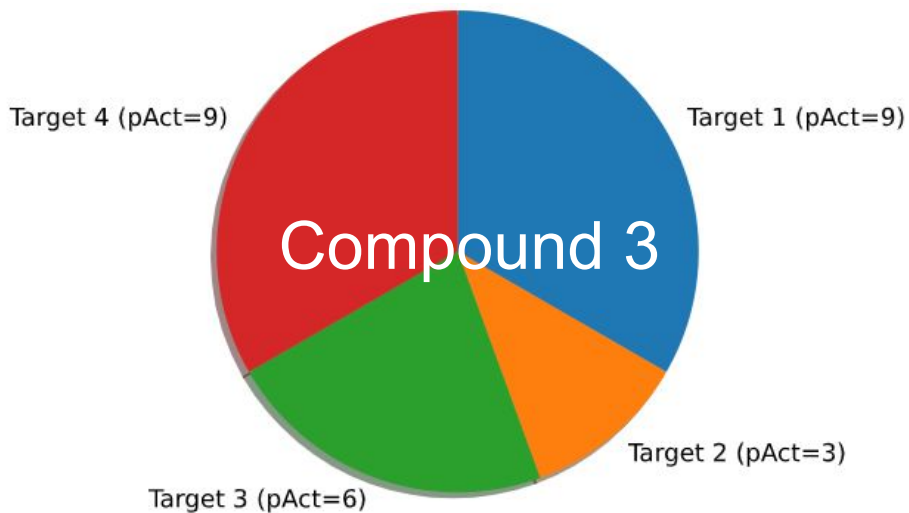
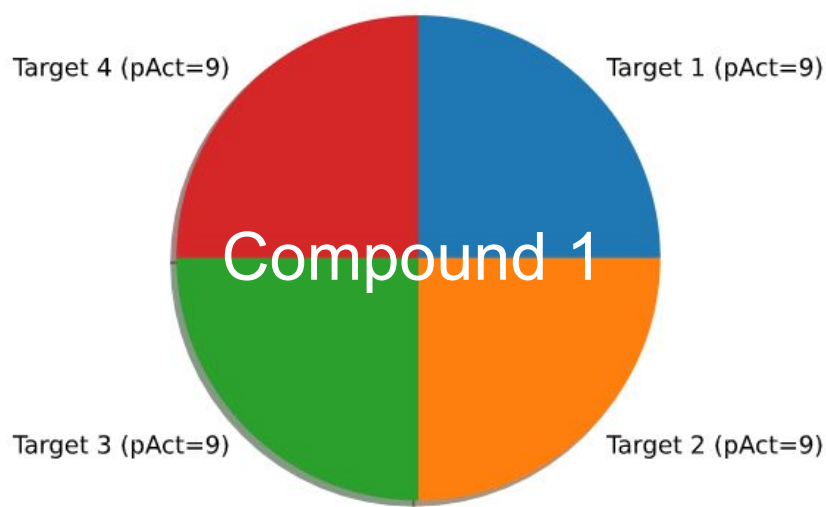
[Visualization of ChEMBL](#)
(version 35; Dec 2024)

Discussion

1. Why do we care selecting *representative, potent, and selective* compounds?
2. How to define following terms mathematically ...
 - a. Representativity?
 - b. Potency?
 - c. Selectivity?

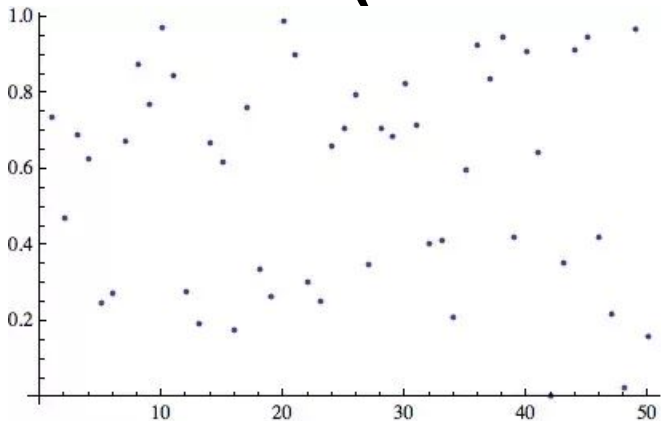


A toy example about how to quantify a compound's potency and selectivity

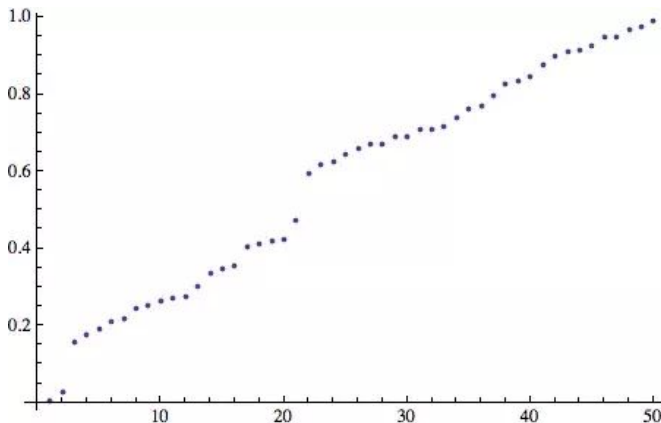


The Gini Index (a.k.a. Gini Coefficient)

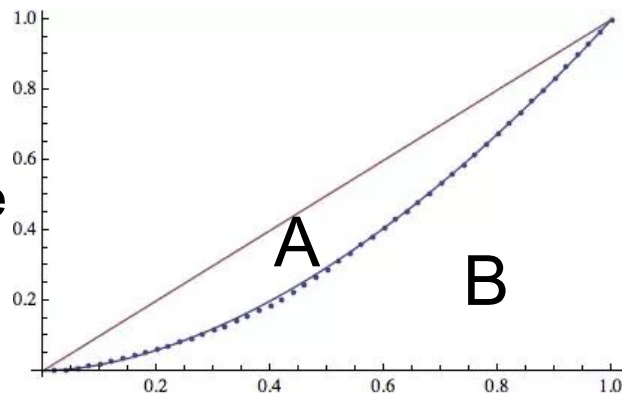
A random
vector of
50 values



Sorted
from low
to high



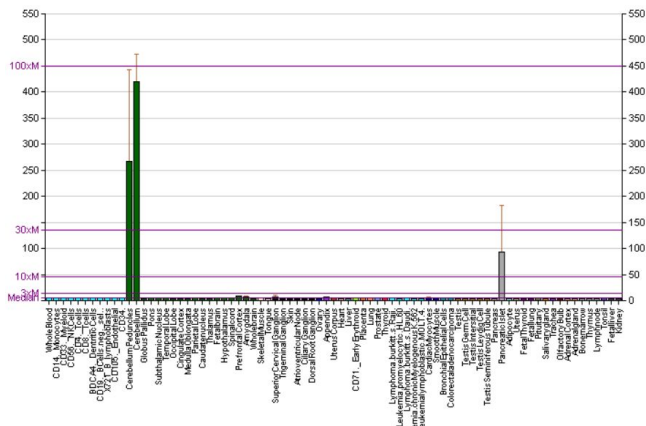
The Gini
Index is
calculated
based on the
cumulative
distribution:
larger
value → large
inequality



$$G = A / (A + B)$$

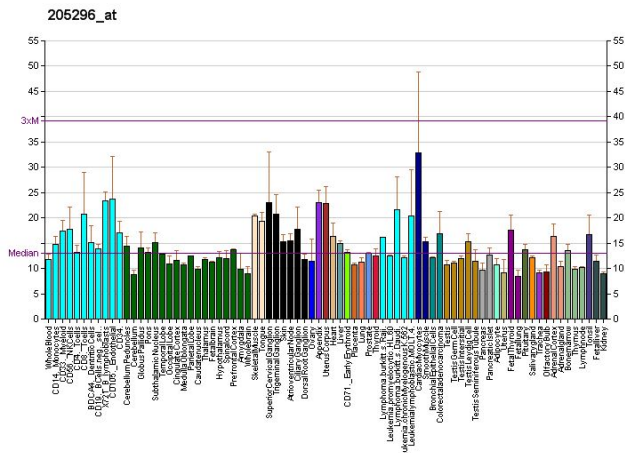
The Gini Index quantifies inequality/ selectivity

NEUROD1

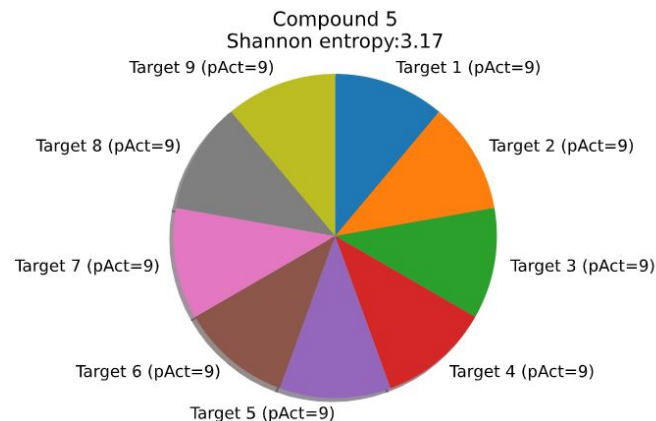
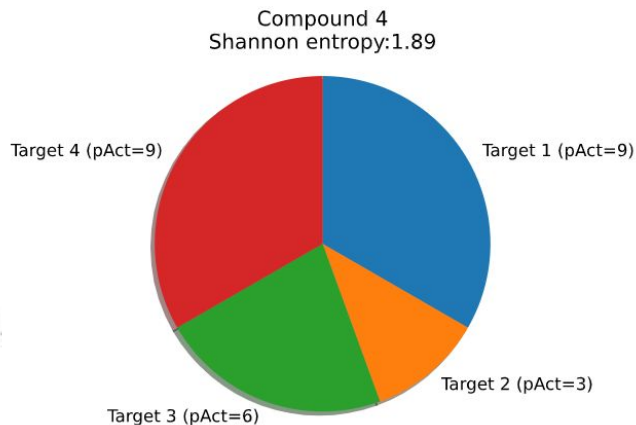
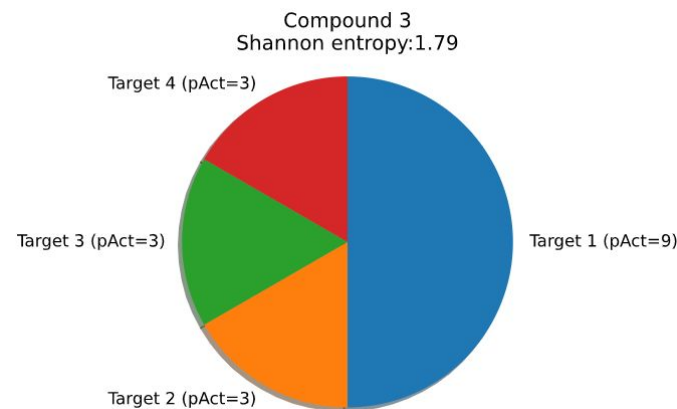
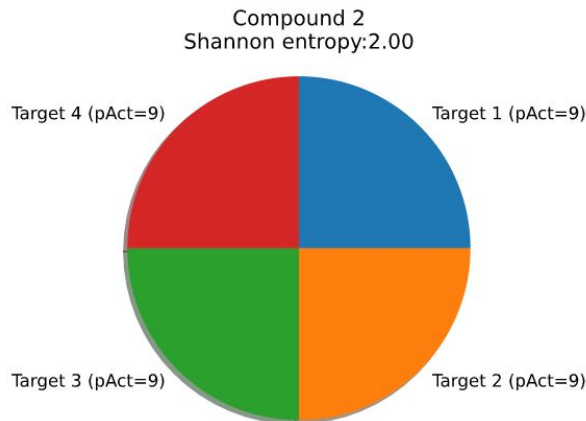
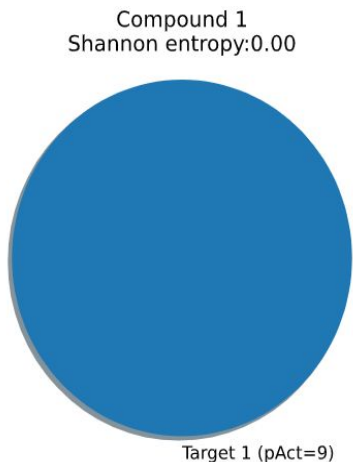


The Gini Index of expression of *NEUROD1* across tissues is near 1, whereas that of *RBL1* is near 0.

RBL1

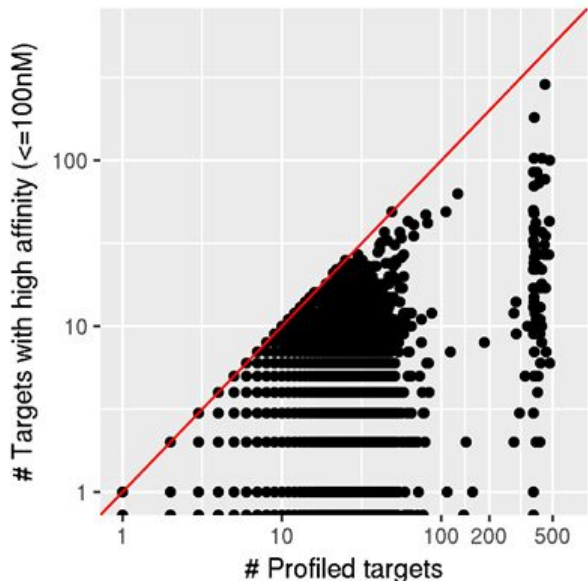


An alternative metric: Shannon's Entropy

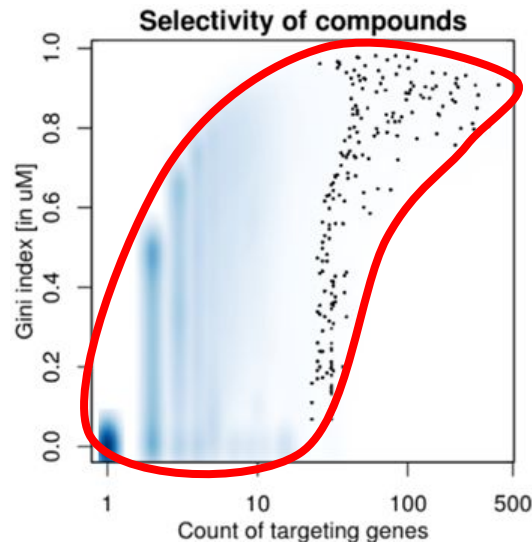


$$H(X) := - \sum_{x \in \mathcal{X}} p(x) \log p(x)$$

Count of targets and selectivity of ChEMBL molecules

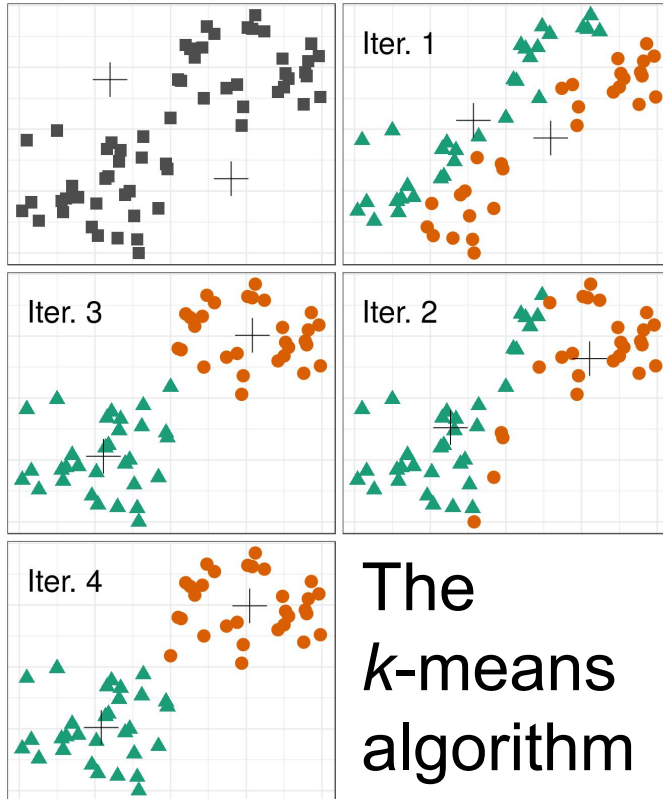


With some exceptions, most compounds are profiled against <100 targets. We distinguish between specific and pleiotropic compounds.



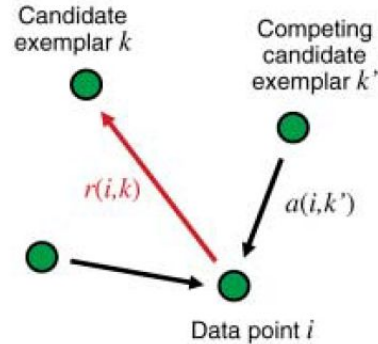
The **shark-fin shape** curve suggests that frequently profiled compounds tend to be more selective (and *vice versa*).

Unsupervised clustering



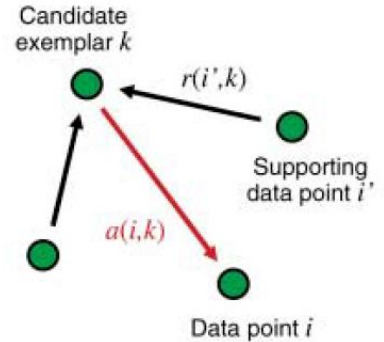
B

Sending responsibilities



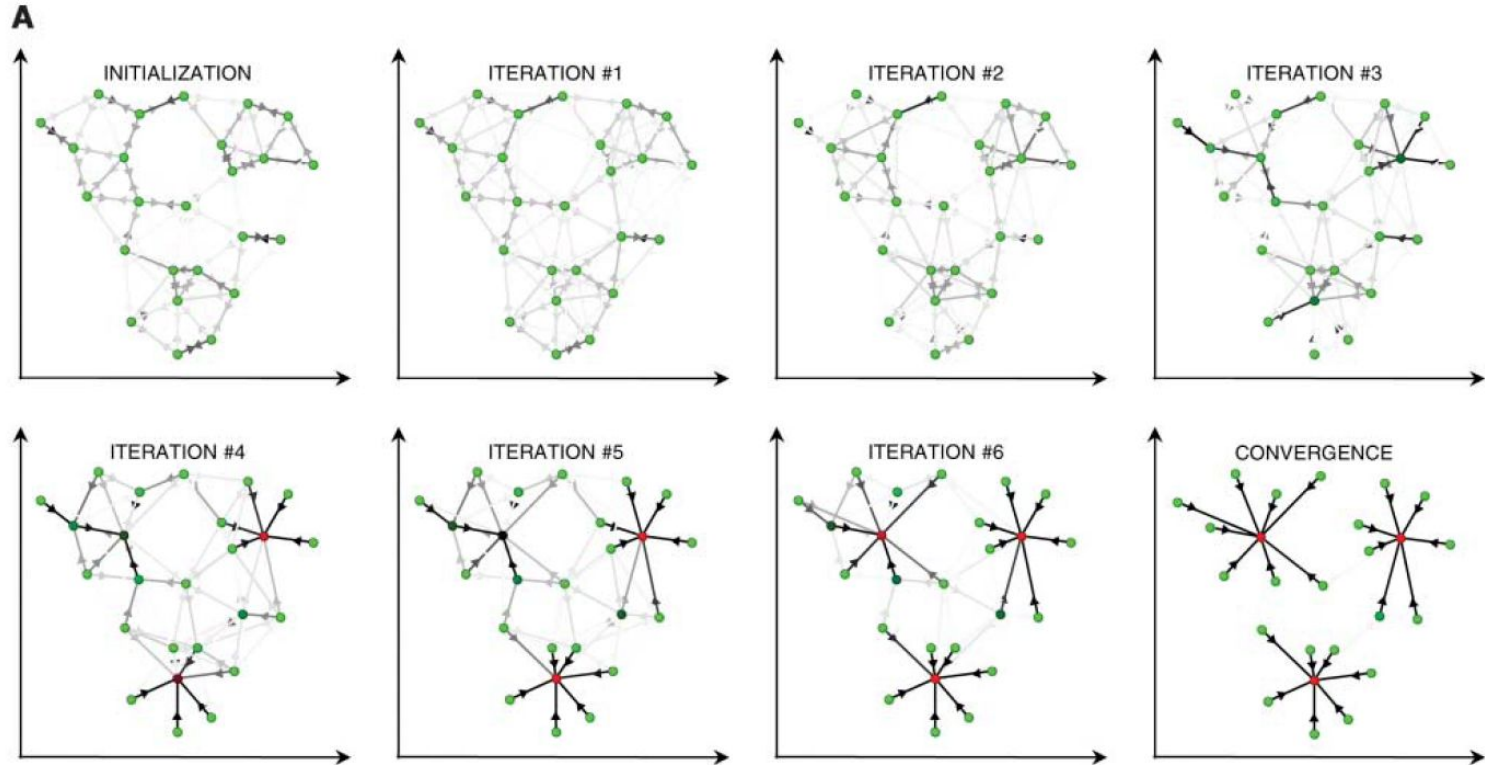
C

Sending availabilities



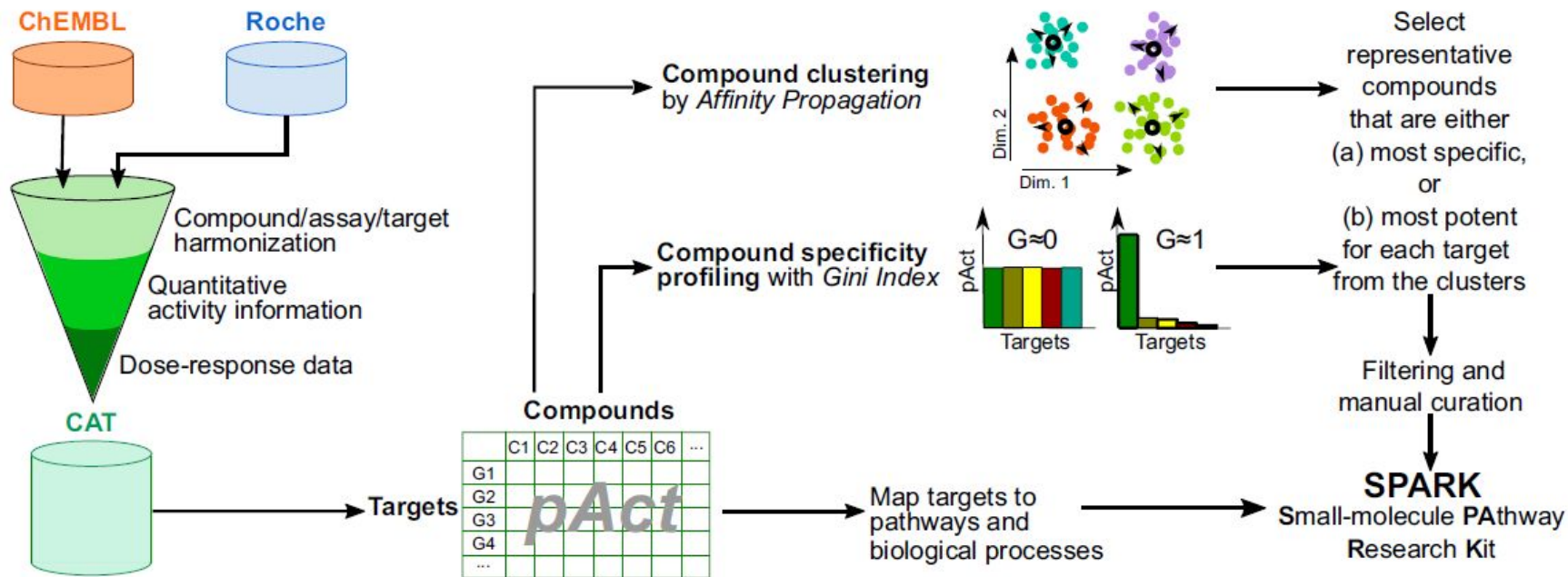
Affinity Propagation updates **responsibilities** and **availabilities** iteratively

Affinity Propagation in action



[A movie of iterations](#)

Construction of SPARK in detail



Harmonization

... of public and Roche internal data

Machine learning

... to select compounds

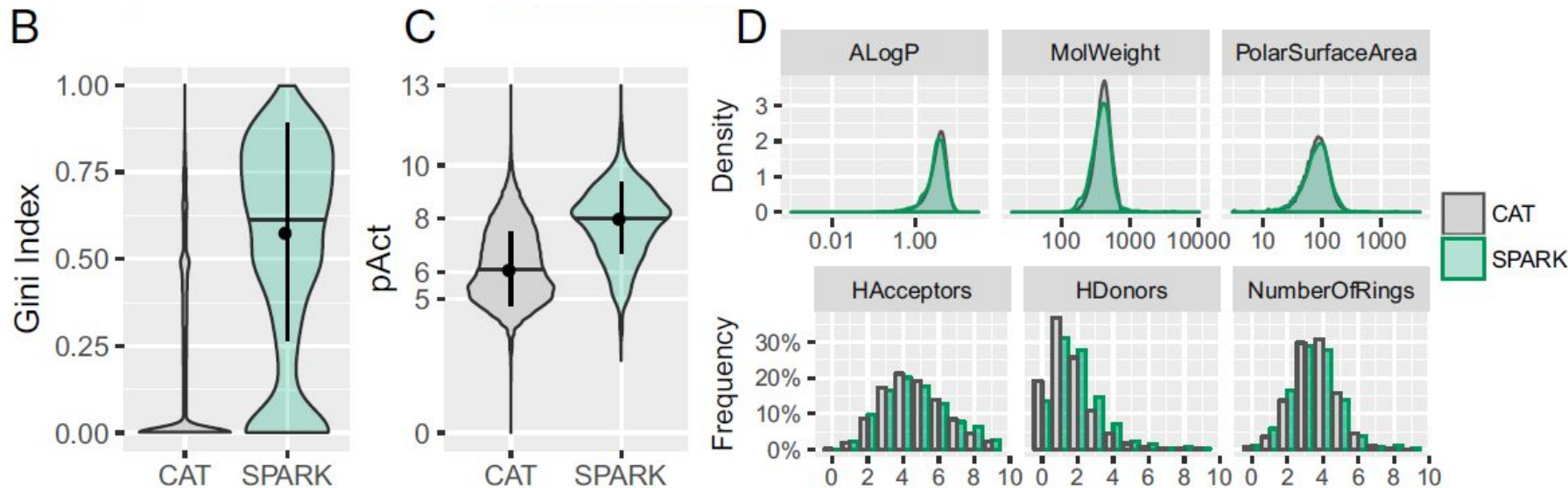
Pathways

... mapped to compounds

Curation

... to enrich quality compounds

SPARK covers the chemical space evenly with representative, potent, and specific compounds



Roudnicky *et al.*, PNAS, 2020,
<https://www.pnas.org/content/early/2020/08/04/1911532117>

Phenotypic screenings by agent and readout

Agent

High-throughput screening
libraries ($\geq 10^6$ molecules)

Genetic libraries ($\sim 10^4$)

Natural products and chemo-
genomic libraries ($\sim 10^3$)

Custom libraries ($\sim 10^0 - 10^2$)

**Boundary of
Feasibility**

Reporters

Gene
expression

Cellular
morphology

Organ/tissue
phenotype

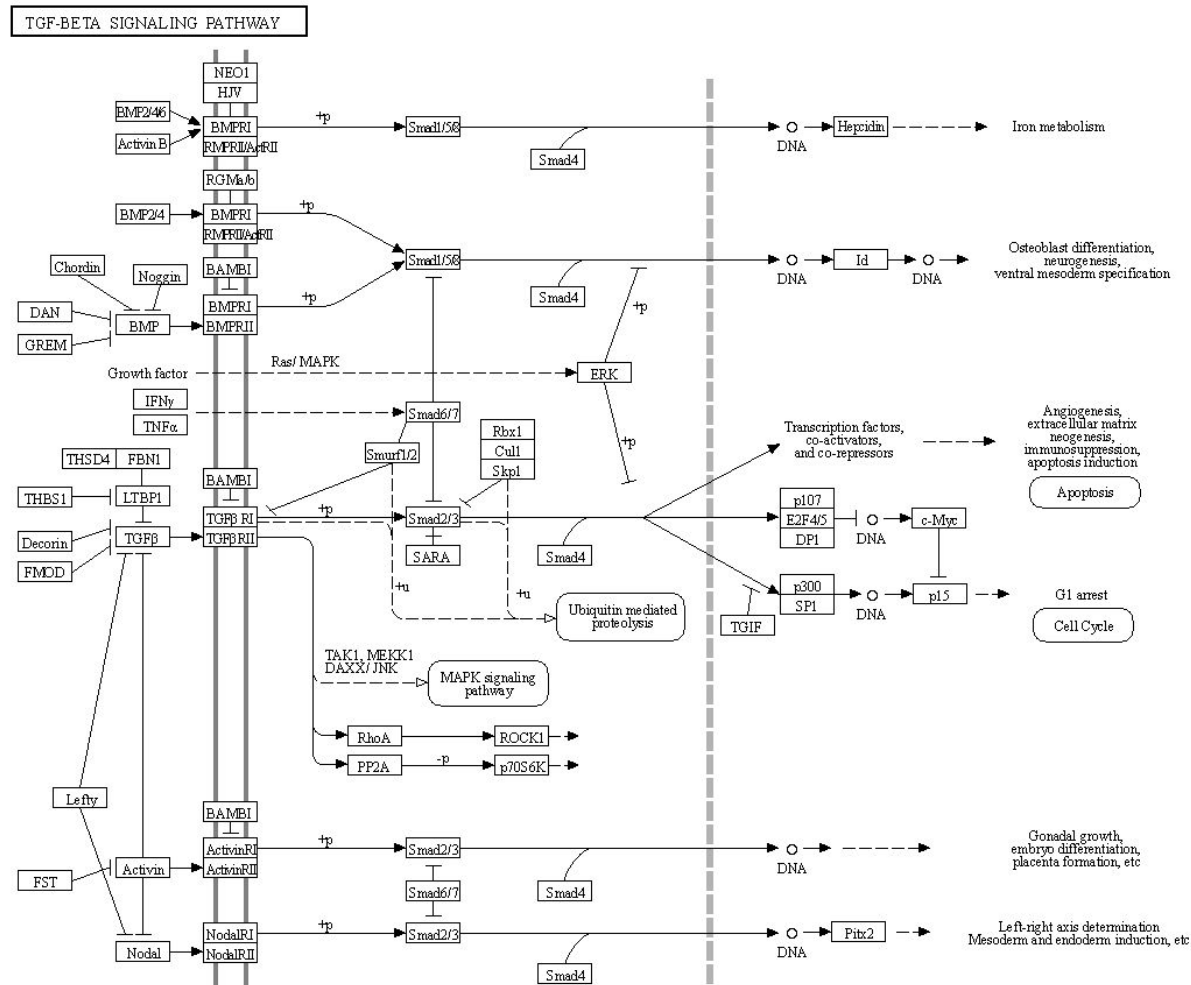
Organism
phenotype

Readout

Mapping genes to biological pathways

Option 1: [KEGG pathways](#), with the example of [TGF- \$\beta\$ signaling pathway](#).

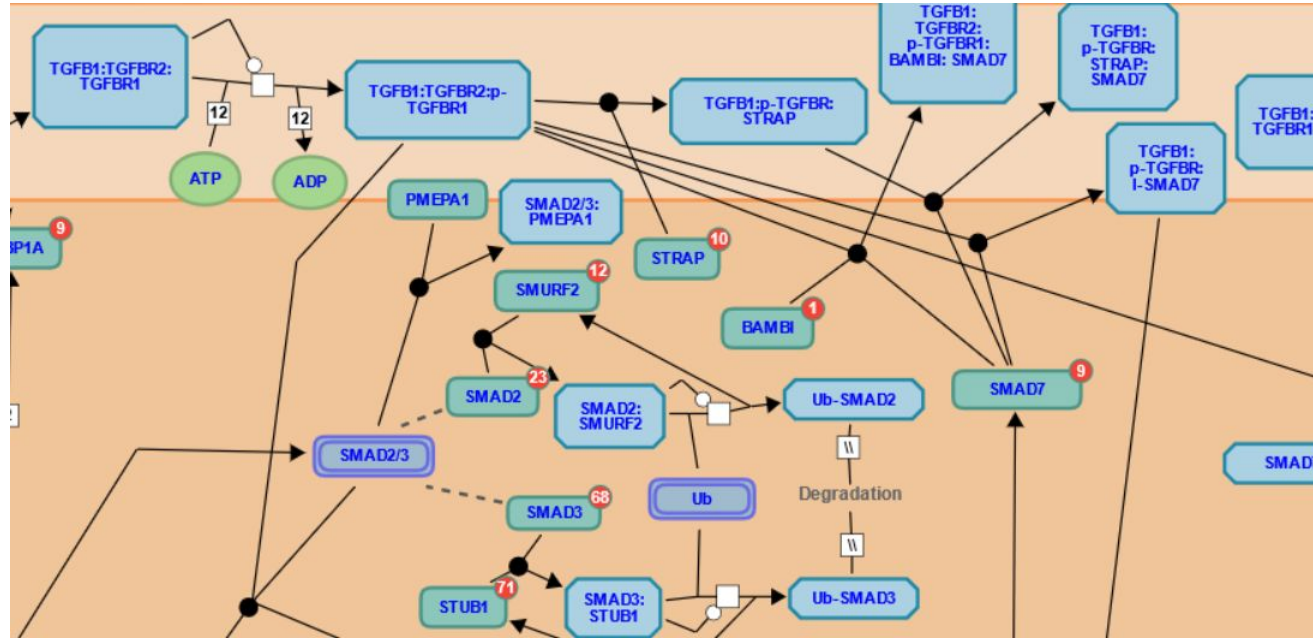
[A RESTful API](#) is available for academic use, with clients in Python and R.



Mapping genes to biological pathways

Option 2: Reactome pathways, with the example of the TGF- β signaling pathway.

Developer's Zone
provides API and
graph database
interfaces.





Mapping genes to biological processes

- Gene Ontology
- UniProtKB keywords
- Example:
TGFR2_HUMAN
(TGF-beta receptor type
-2, P37173)

Keywords

Molecular function	Kinase, Receptor, Serine/threonine-protein kinase, Transferase
Biological process	Apoptosis, Differentiation, Growth regulation
Ligand	ATP-binding, Magnesium, Manganese, Metal-binding, Nucleotide-binding

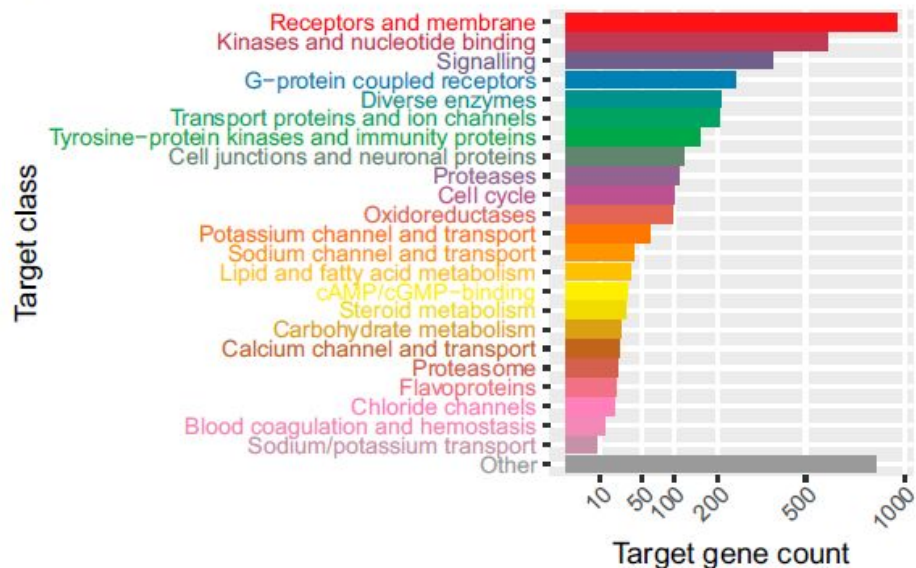
[illegible]

GO - Biological process

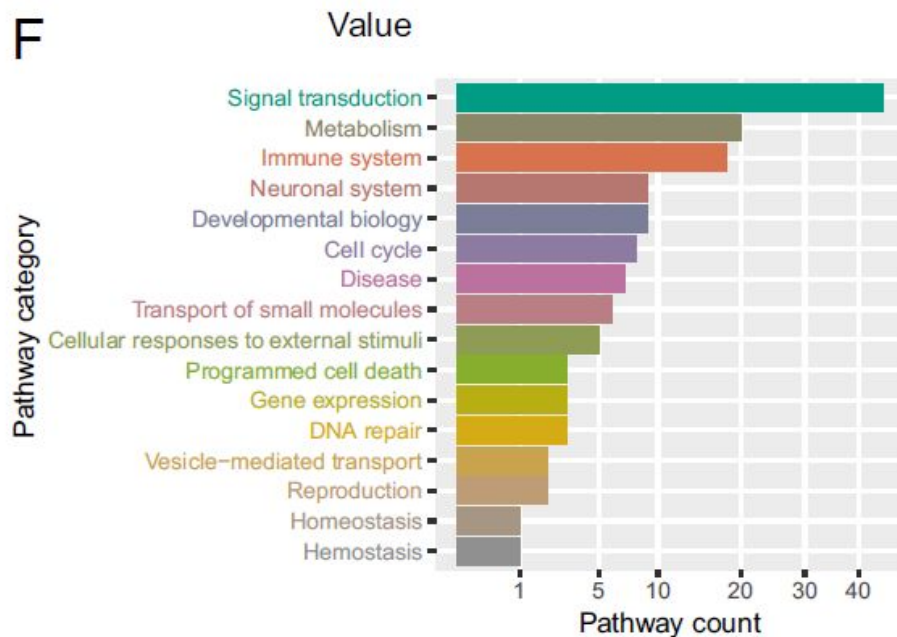
- activation of protein kinase activity Source: BHF-UCL
- aging Source: Ensembl
- animal organ regeneration Source: Ensembl
- apoptotic process Source: UniProtKB
- atrioventricular valve morphogenesis Source: BHF-UCL
- blood vessel development Source: BHF-UCL
- brain development Source: BHF-UCL

SPARK covers the target space evenly with representative, potent, and specific compounds

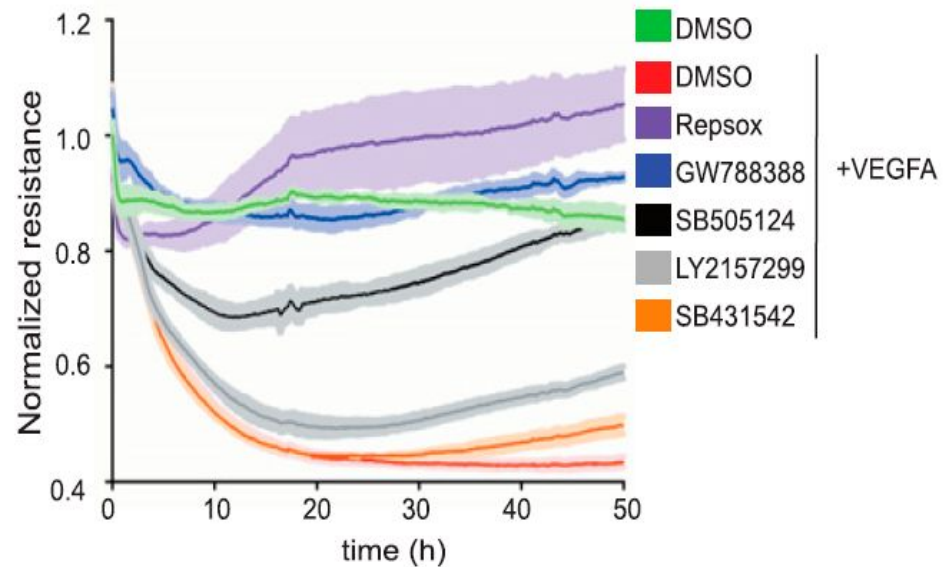
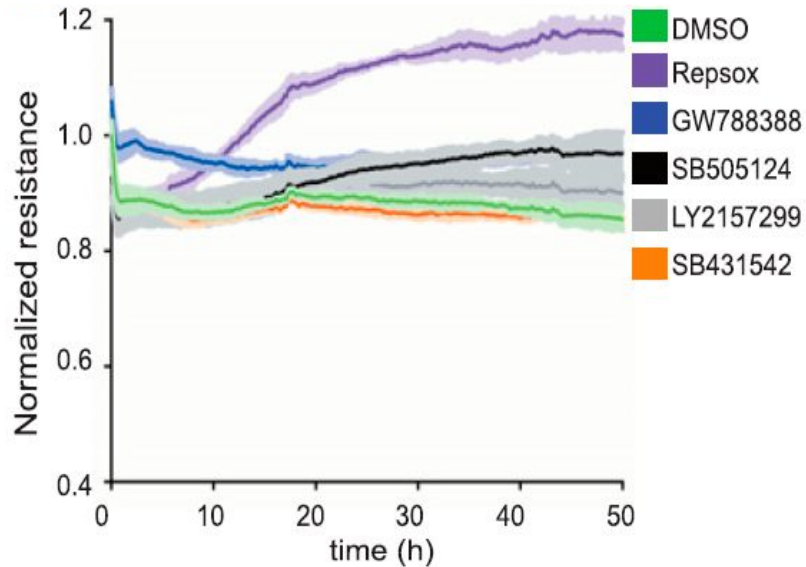
E



F

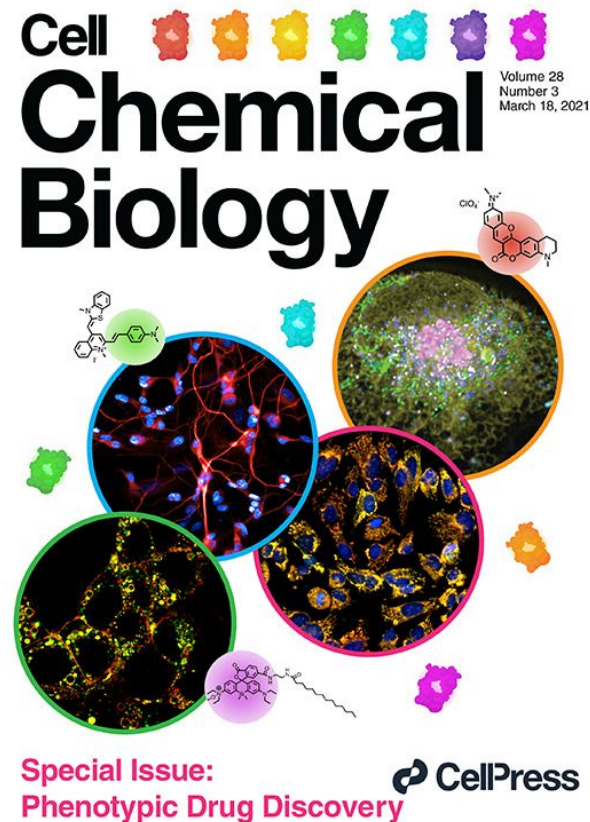


Screening with SPARK in endothelial cells identified TGF- β pathway genes as potential targets for diabetic retinopathy



Conclusions about chemogenomic library

- Phenotypic drug discovery can lead to first-in-class drugs with novel mechanisms;
- Unsupervised machine learning and data modelling contribute to build chemogenomic libraries;
- We can link drug candidates via targets to biological pathways and processes.



The limitations of genetic screening in phenotypic drug discovery

Table 3. Genetic screening limitations and mitigation strategies

Genetic screening limitations	Mitigation strategy
Effects have a delayed onset	Extend assay duration
Effects may not capture allosteric or scaffolding function of a protein	Use genetic libraries that have well-annotated small molecule tool compounds or antibody/protein tools, development of tunable KD protocols,
Genetic redundancy	Develop combinatorial gene editing protocols to examine contribution of multiple genes to a phenotype
Workflows for genome-wide screens are not compatible with use of translatable cellular models of disease	Use shorter and focused gene lists leveraging human genetics, omics, literature, and AI predictions
Resource and time intensive	Use of “generic” assay readouts (e.g., cell painting, transcriptomics)
Focus on gene KD/KO may miss activation mechanisms	Additional CRISPRa or cDNA screen
May miss selective modulation via macromolecular complexes	N/A

KD, knockdown; KO, knockout; N/A, not applicable.

The limitations of small molecule screening in phenotypic drug discovery

Table 2. Small molecule screening limitations and mitigation strategies

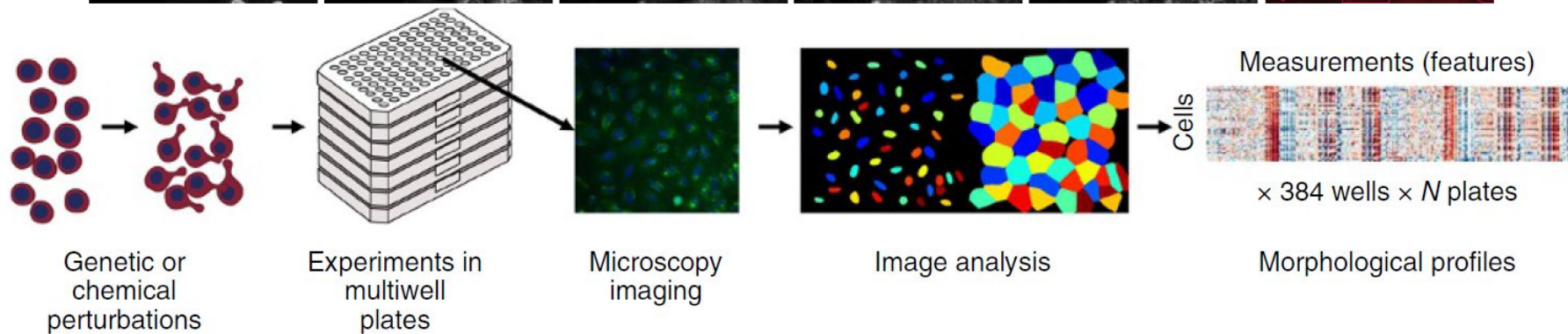
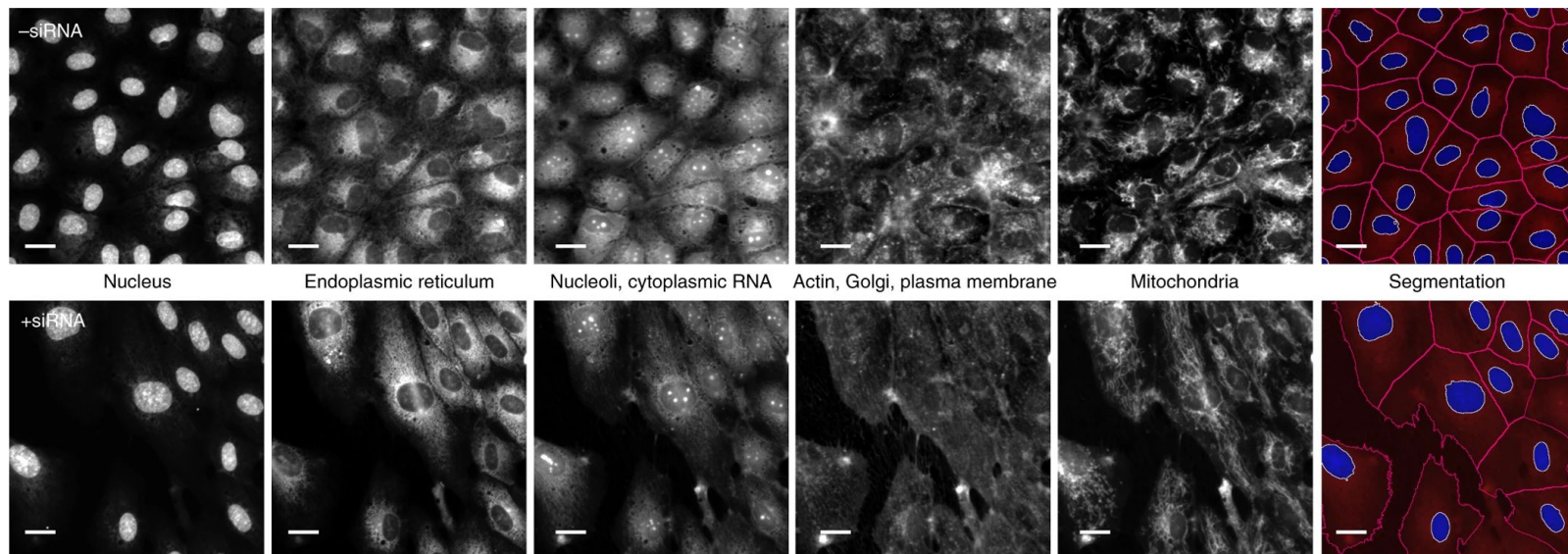
Small molecule screening limitations	Mitigation strategies
Chemogenomics libraries only cover ~5%–10% of the genome	Additional genome coverage via secreted proteins or receptor antibodies
Large legacy compound libraries only cover a fraction of the genome	New compound classes address a larger fraction of the genome (covalent, photoactivatable, and molecular glue compounds)
Limited throughput for complex assays, especially <i>in vivo</i> assays	Screening library compression, smaller compounds (e.g., fragments), biological diversity-driven libraries
Frequent lack of selectivity for related protein homologs	New compound classes may offer greater selectivity (covalent and molecular glue compounds)
Potential species translation hurdle	Additional species phenotypic assays, 10× <i>in vivo</i> dosing strategy, complex human models
Target and MoA deconvolution post-screen often challenging and may be required for successful outcome	New compound classes (covalent, photoactivatable, and glue) may facilitate target and MoA deconvolution through proteomics

Offline activities of Module II

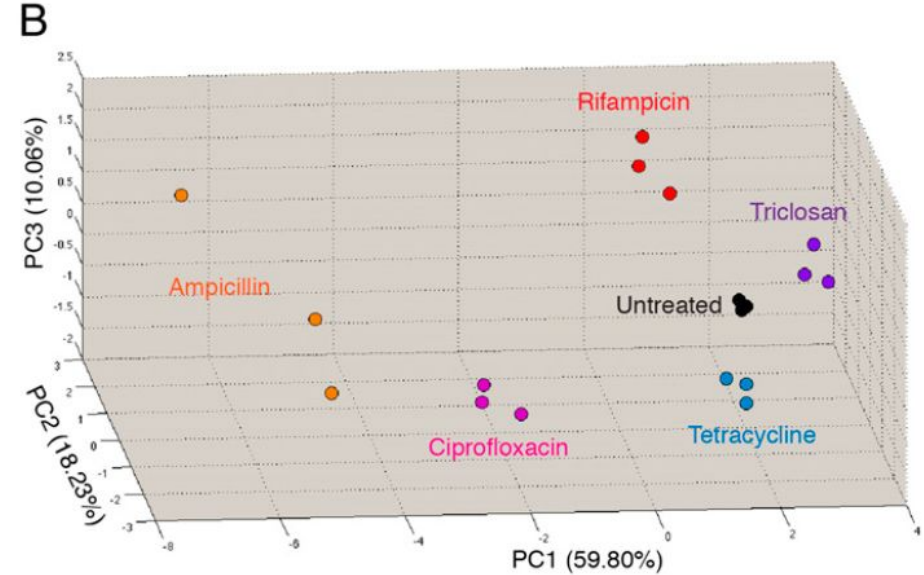
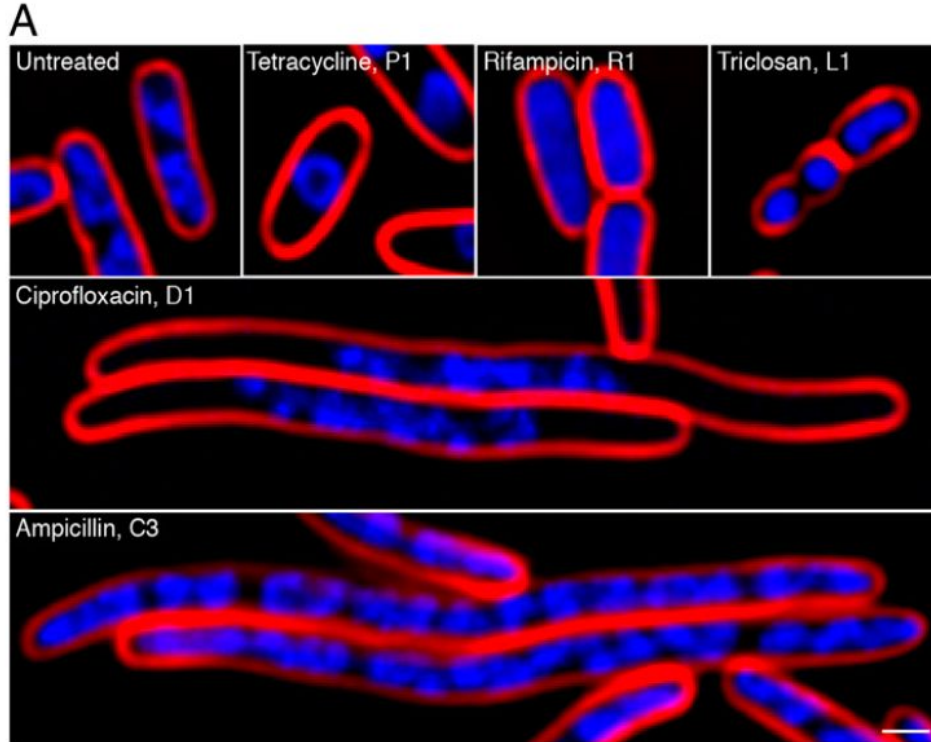
Please use your favourite programming language (shell scripts, python, R, for instance) and APIs (Application Programming Interfaces) of databases to perform following operations. Submit your code.

1. Retrieve all approved drugs from the ChEMBL database, sort them by approval year and name ([a Python example is here](#); documentations of the ChEMBL API can be found [here](#));
2. For each approved drug **since 2019** that you identified in step (1), retrieve a list of UniProt accession numbers, namely protein targets associated with the drug;
3. For each protein with a UniProt accession number that you identified in step (2), retrieve UniProt keywords associated with it. [You can use the UniProt API, documented here](#). [Python](#) and [R](#) clients are also available.

Morphology as screening readout



Cytological profiling for antibiotics discovery



P: Protein translation inhibitors

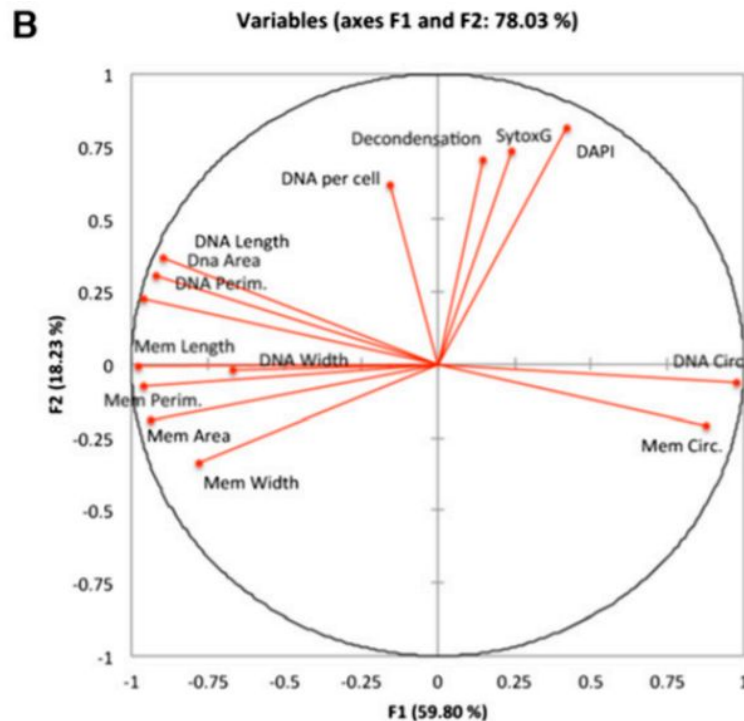
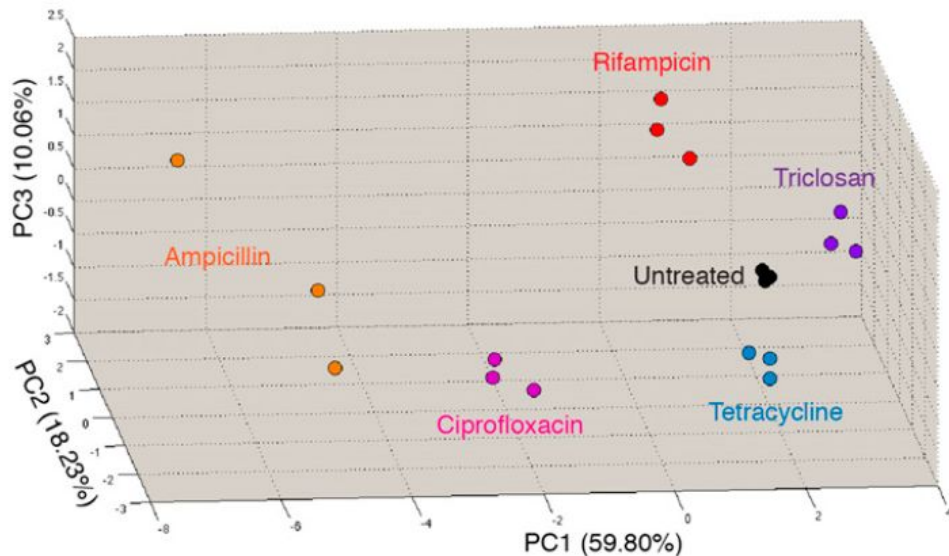
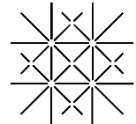
R: RNA transcription inhibitors

D: DNA replication inhibitors

L: Lipid biosynthesis inhibitors

C: Cell-wall synthesis inhibitors (peptidoglycan)

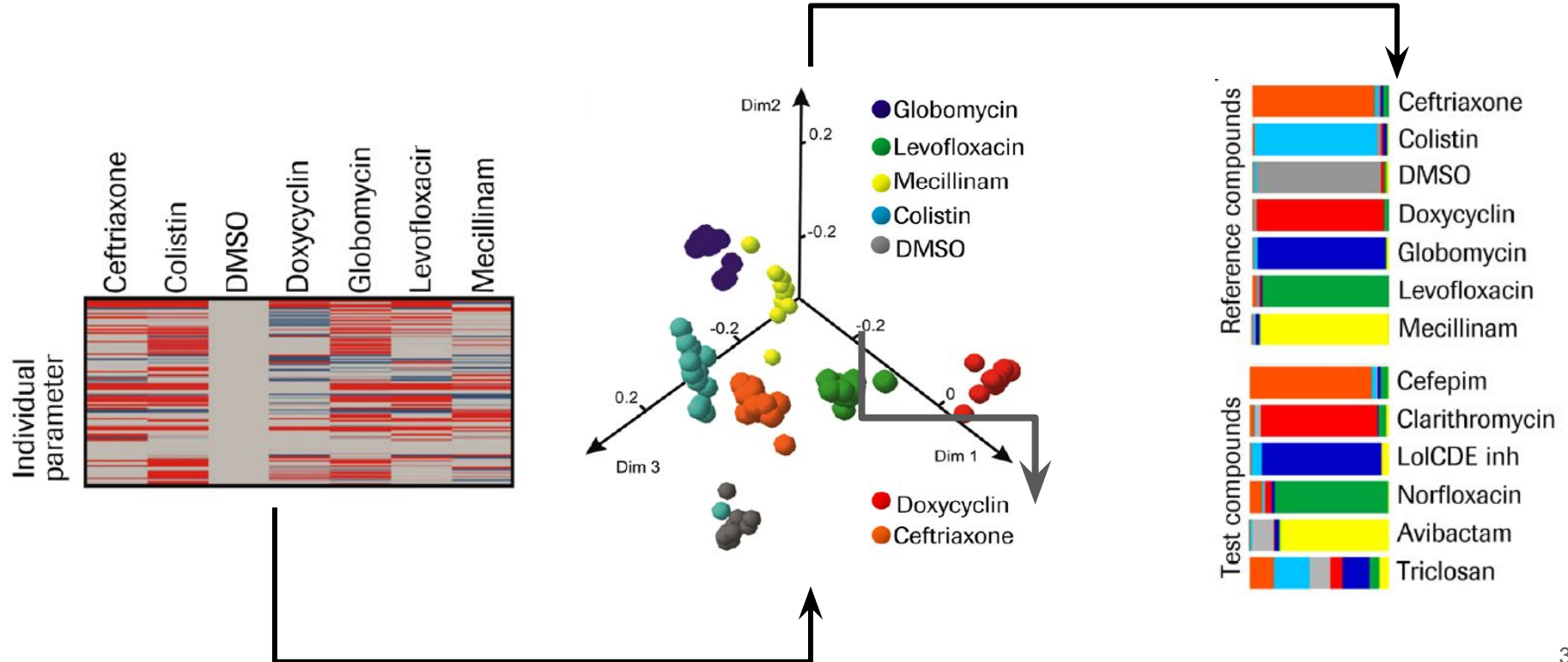
Principal components are linear combination of morphological features



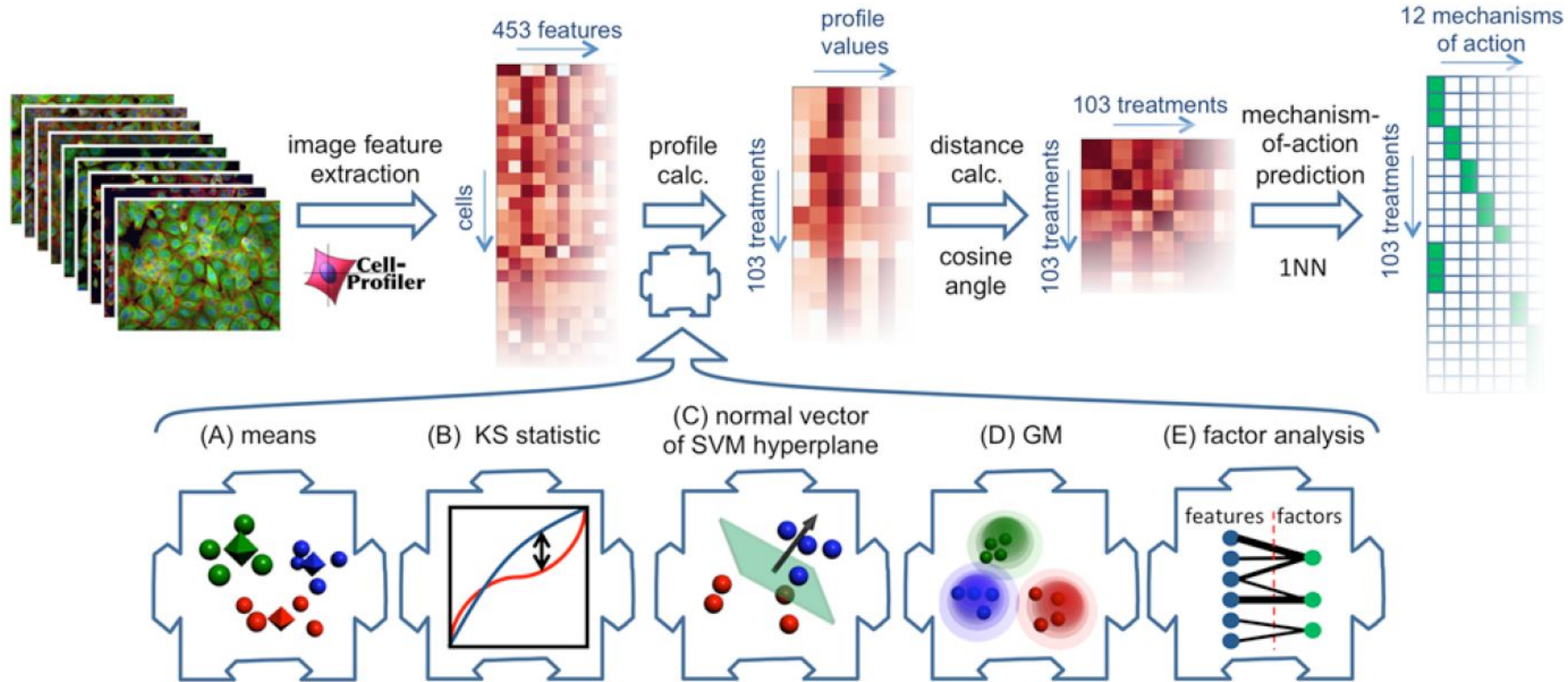
Membrane area, μm^2 DNA area, μm^2 Membrane perimeter, μm DNA perimeter, μm Membrane length, μm DNA length, μm No. of nucleoids per cell

Membrane width, μm DNA width, μm Membrane circularity DNA circularity SytoxG intensity DAPI intensity Decondensation

Morphology classifies compounds by MoA



Comparison of computational methods



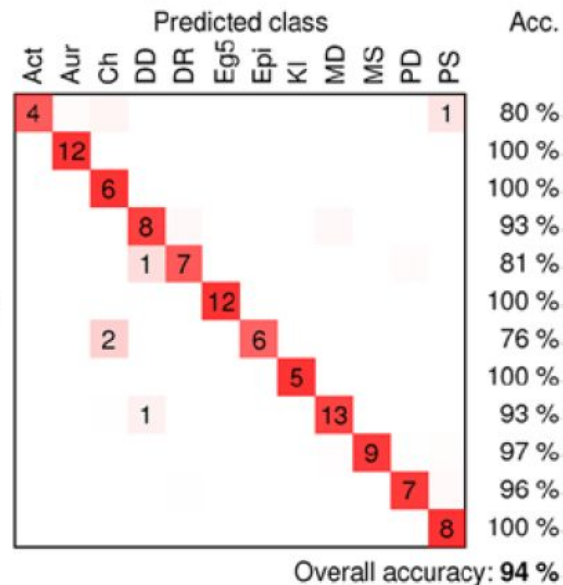
Do the benchmark and use Occam's Razor

Table 1. Accuracies for classifying compound treatments into mechanisms of action.

Method	Accuracy, %
Means	83
KS statistic	83
Normal vector to support-vector machine hyperplane	81
With recursive feature elimination	64
Distribution over Gaussian mixture components	83
Factor analysis + means	94

True mechanistic class

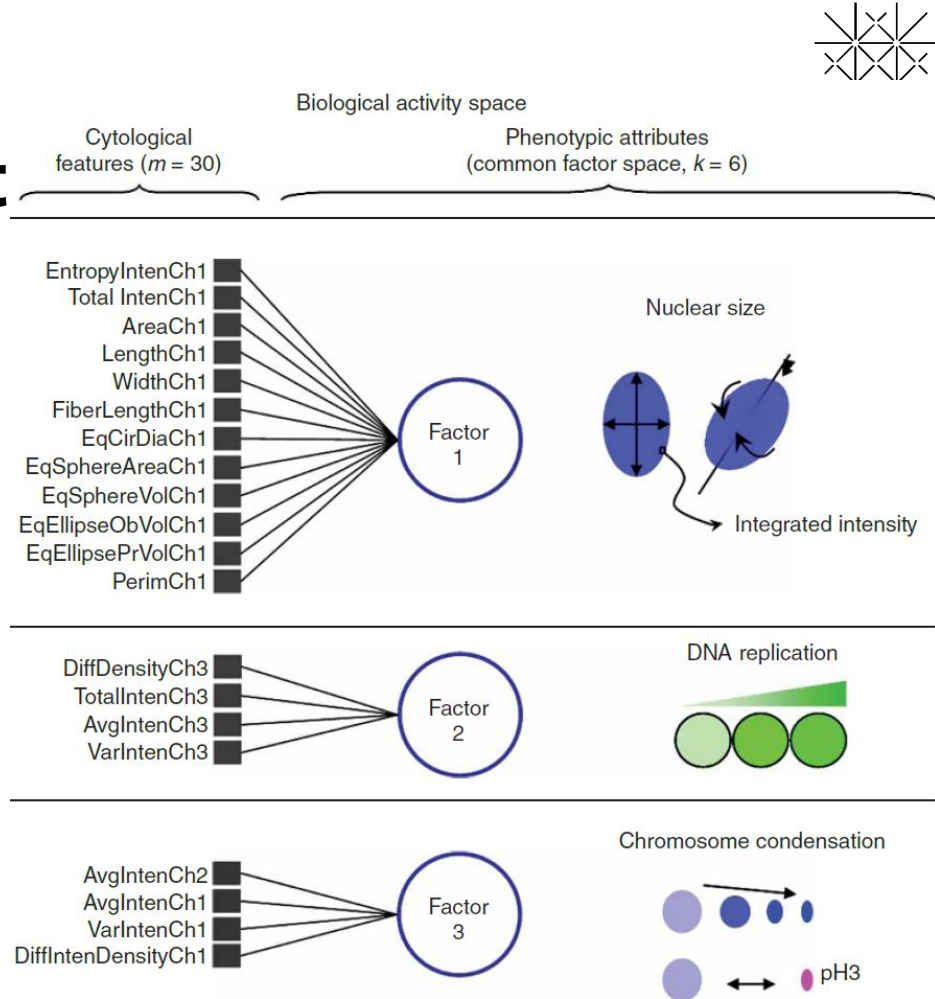
Actin disruptors	Act
Aurora kinase inhibitors	Aur
Cholesterol-lowering	Ch
DNA damage	DD
DNA replication	DR
Eg5 inhibitors	Eg5
Epithelial	Epi
Kinase inhibitors	KI
Microtubule destabilizers	MD
Microtubule stabilizers	MS
Protein degradation	PD
Protein synthesis	PS



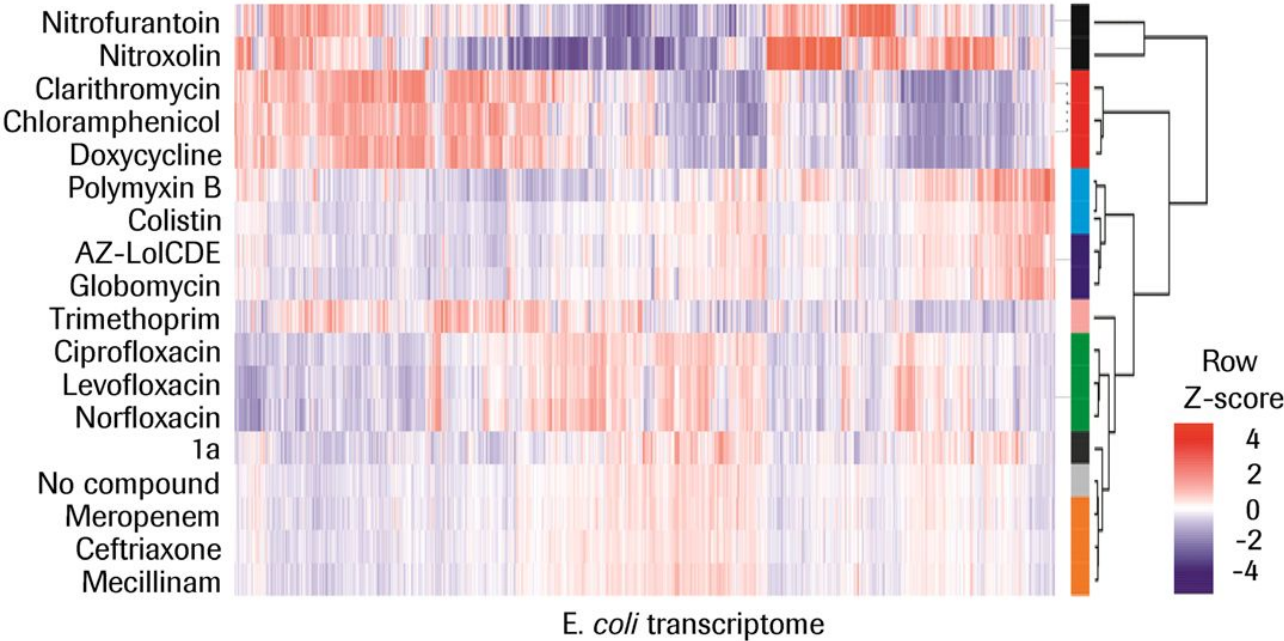
A possible explanation for the success of latent variable models

$$\begin{matrix} \text{Cells} \\ \begin{pmatrix} x_{11} & \dots & x_{1m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{nm} \end{pmatrix} \\ \text{Cytological features} \end{matrix} = X_{nm} = \underbrace{\sum_{i=1}^k L_{ni} F_{im}}_{k\text{-factor space}} + \varepsilon_{nm}$$

A common latent factor model



Morphology and gene expression used jointly

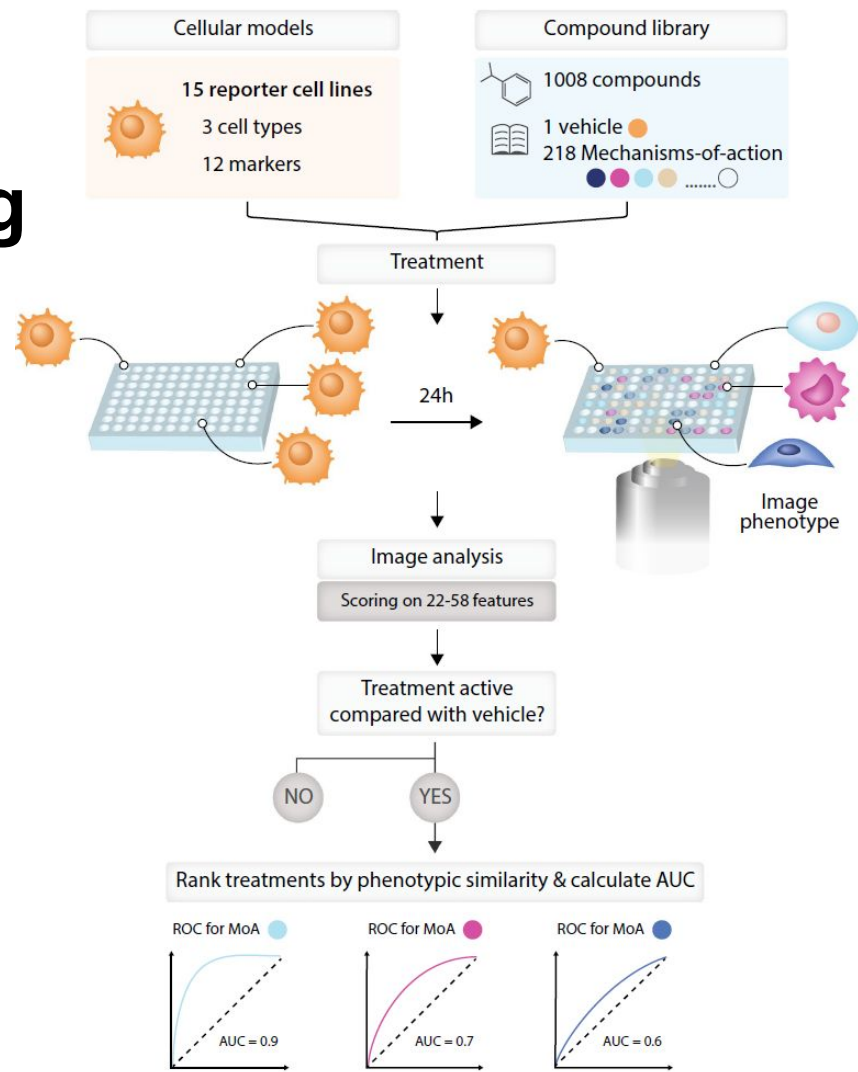
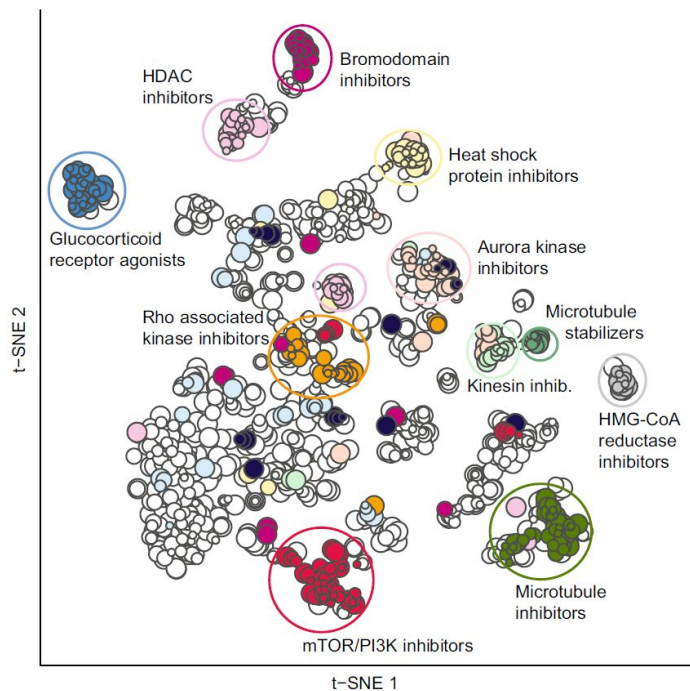


Gene-set
enrichment
analysis

Reporter
assays

Pathway-
Phenotype
associations

A multi-cell-type, 1008-compound screening by Cox *et al.* (2020)



Conclusions

- Chemogenomic library and image-based profiling are useful tools for phenotypic screening.
- Integration of biological knowledge, high-throughput data, and statistical modelling empowers phenotypic drug discovery.

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Connect the lines!

- Phenotypic screening
- Modified natural products
- Biologics
- Target-based screening

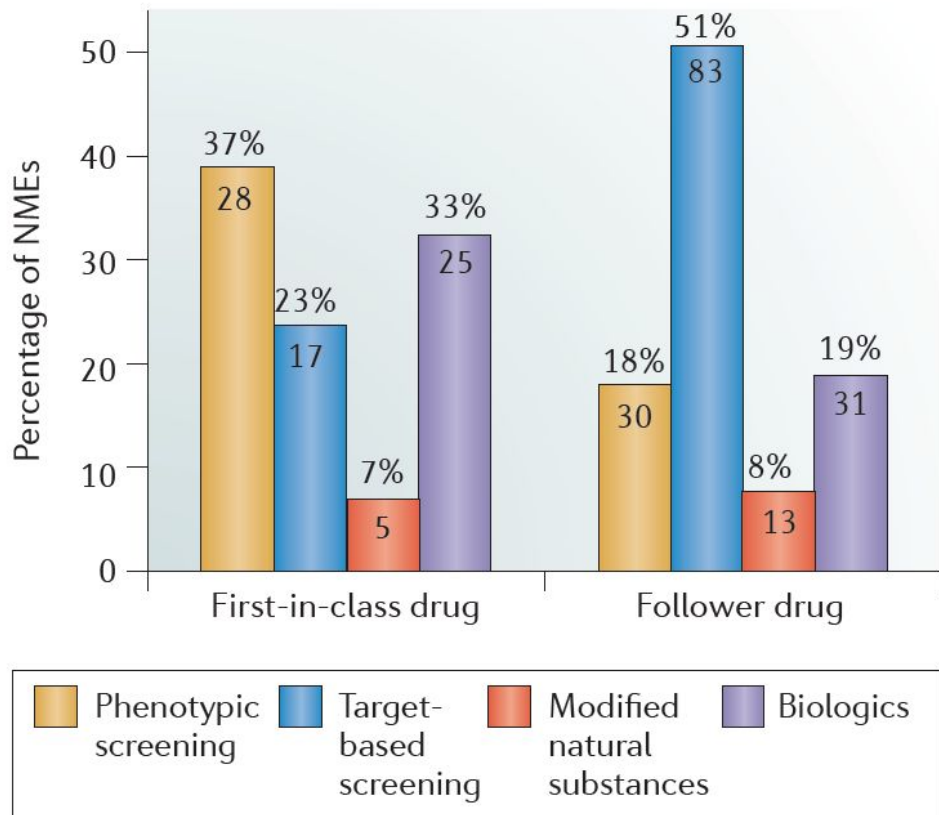
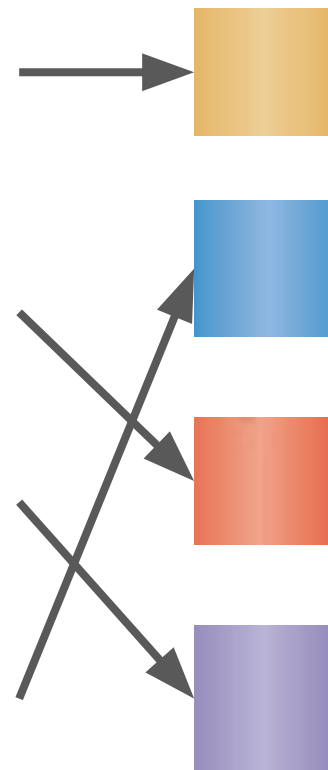
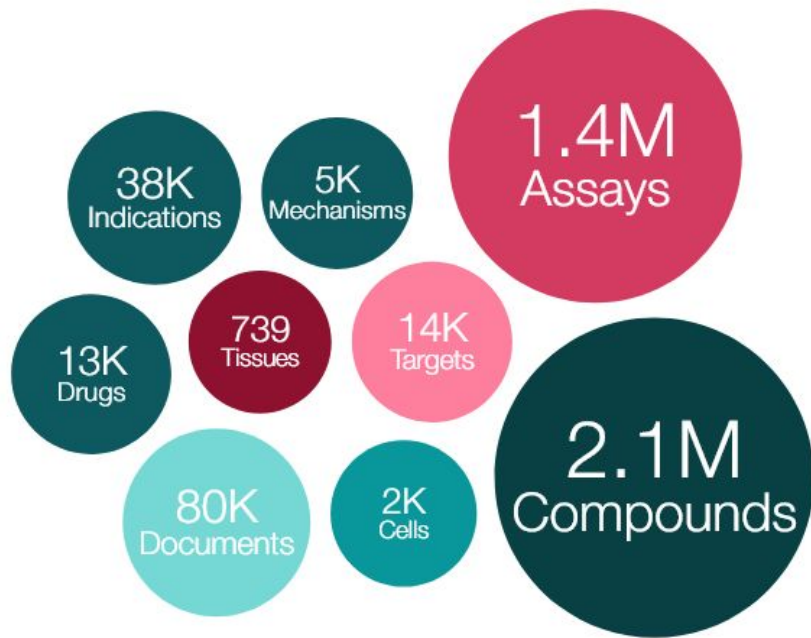
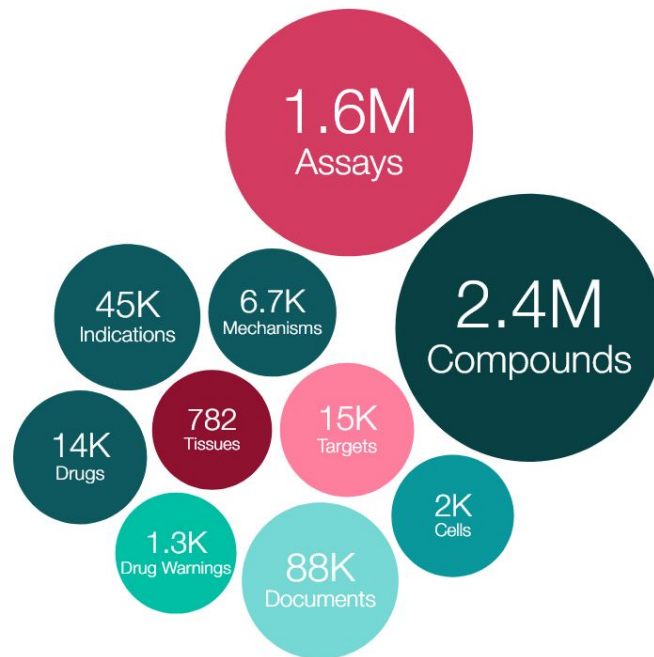


Figure 2 | The distribution of new drugs discovered between 1999 and 2008, according to the discovery strategy.

The evolution of ChEMBL database

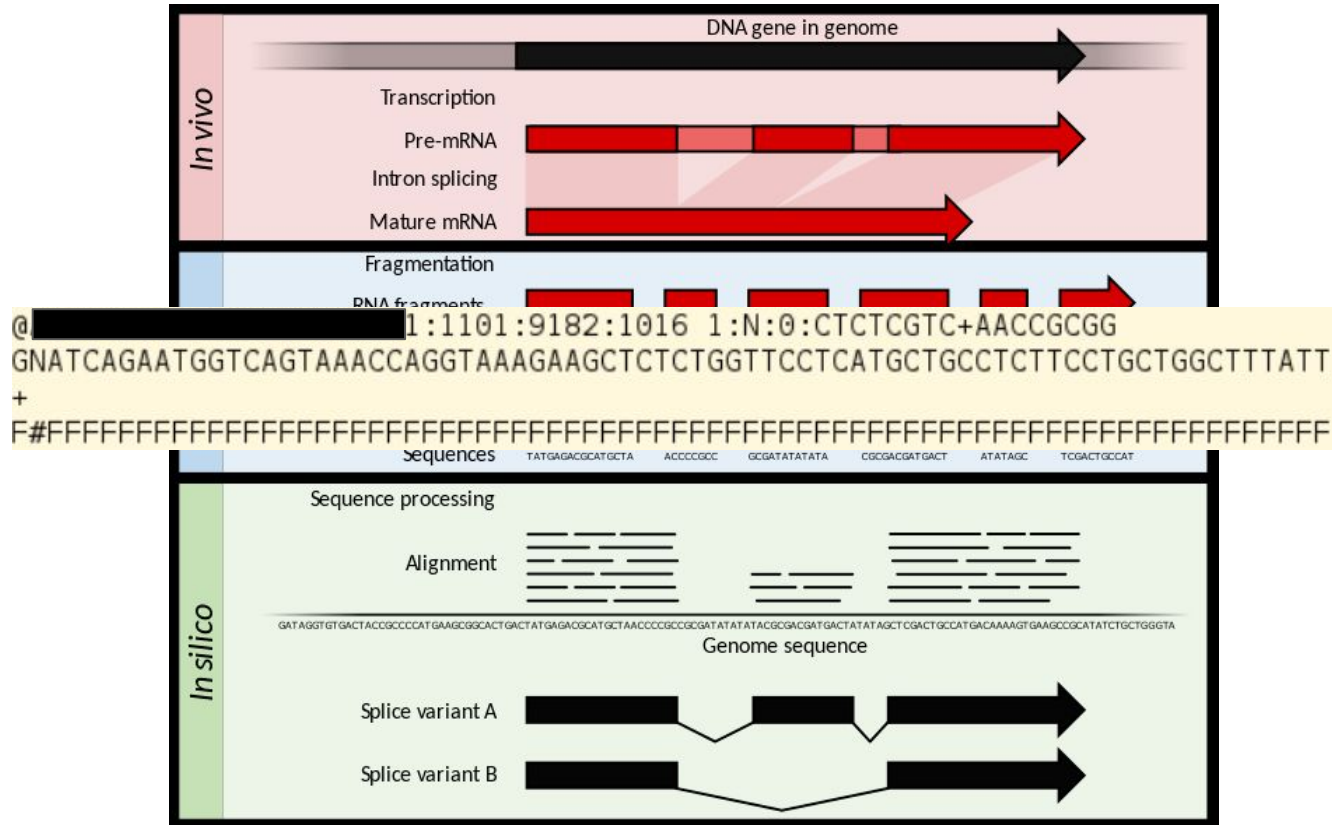


Visualization of ChEMBL (2021)

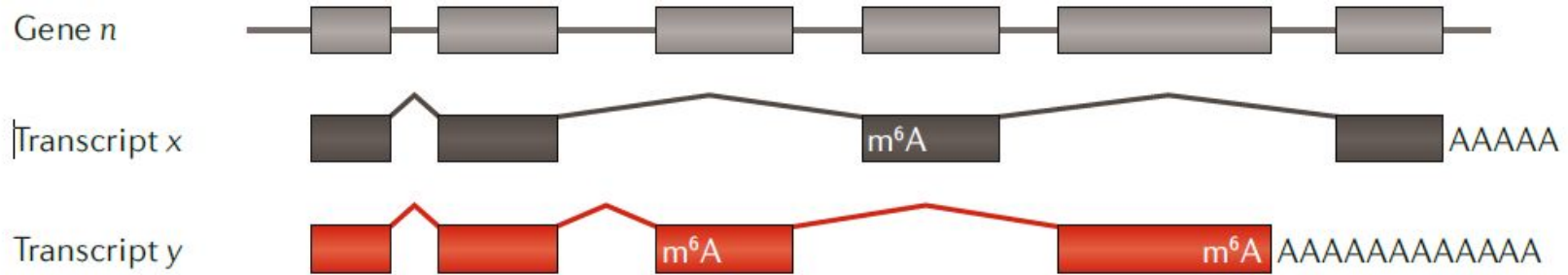


Visualization of ChEMBL
(version 33; 2024)

Transcriptome profiling by RNA sequencing



Transcriptome profiling by RNA sequencing



Ambiguous
to exon



Unambiguous
to exon



Ambiguous
to isoform



Unambiguous
to isoform

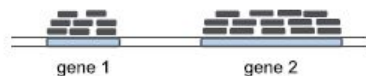


Read Mapping

Count collection

Normalization by library size

Differential Gene Expression Analysis



	sample A1	sample A2	sample B1	sample B2
gene 1	8	10	100	200
gene 2	14	15	15	40
gene 3	33	40	35	70
...	...	...	...	...
gene N	100	120	105	220

	sample A1	sample A2	sample B1	sample B2
gene 1	8	10	100	200
gene 2	14	15	115	40
gene 3	33	40	35	70
...	...	...	...	...
gene N	100	120	105	220

Tot. reads:
5 millions

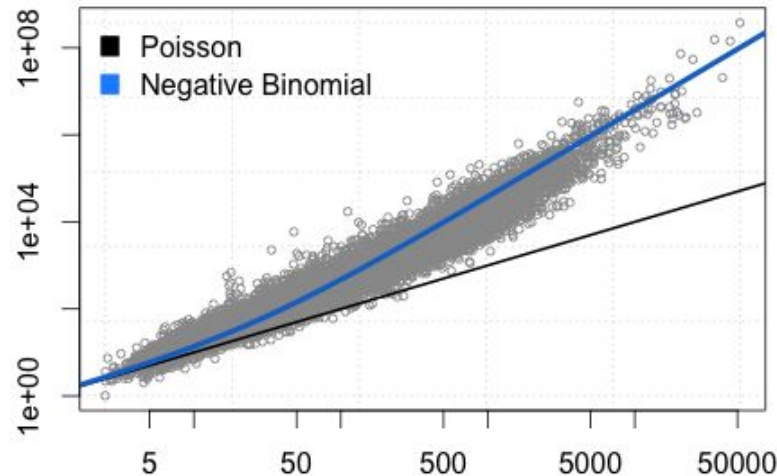
Tot. reads:
10 millions

	sample A1	sample A2	sample B1	sample B2
gene 1	0.16	0.20	2.00	2.00
gene 2	0.28	0.30	0.30	0.40
gene 3	0.66	0.80	0.70	0.70
...	...	...	...	...
gene N	2.00	2.40	2.10	2.20

Differential gene expression



Pooled gene-level variance (log10 scale)



Mean gene expression level (log10 scale)

Tools: *edgeR* and *DESeq2*

Probability theory and statistical tools discussed

- Distributions
 - Gaussian distribution (used in linear model)
 - Bernoulli distribution $\rightarrow$ Binomial distribution $\rightarrow$ Negative binomial distribution
 - Poisson distribution $\rightarrow$ Negative binomial distribution
 - Poisson distribution $\longleftrightarrow$ Exponential distribution
- Statistical methods
 - Bootstrapping method
 - Student's t-test
 - Wilcoxon-Mann-Whitney test
 - Kolmogorov-Smirnov test

Interpret differential gene expression data with gene-set enrichment analysis

Reactome pathways

Gene Ontology

UniProt Keywords

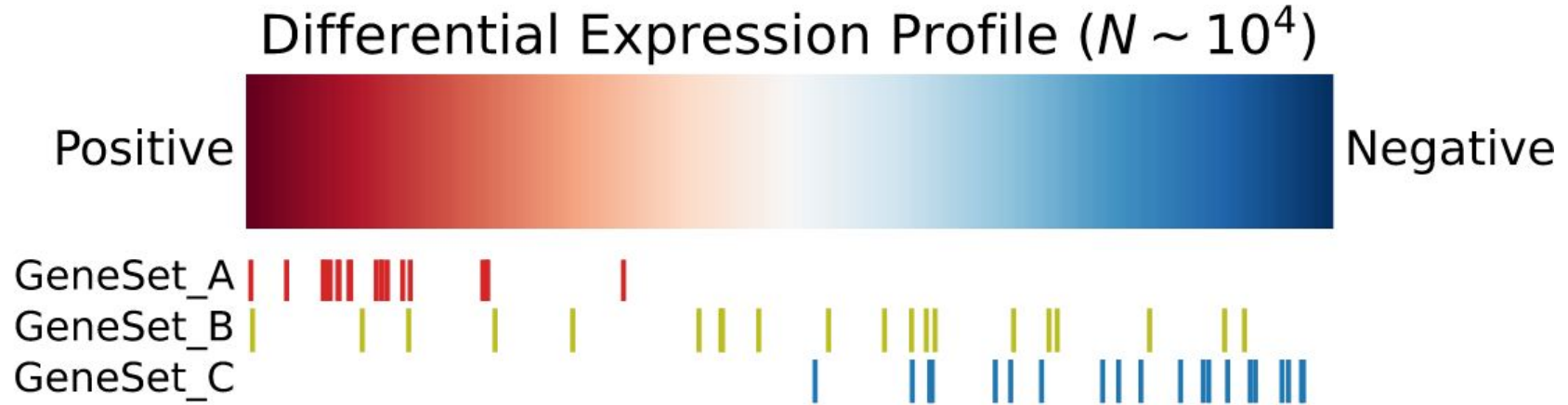
Literature

Gene (N~10 ⁴)	G ₁	G ₂	G ₃	G ₄	G ₅	...	G _{N-3}	G _{N-2}	G _{N-1}	G _N
Change (log2FC)	3.0	2.8	2.5	1.5	1.2	...	-0.8	-1.2	-1.5	-2.2

Differential gene expression results

Gene-set Enrichment Analysis Methods

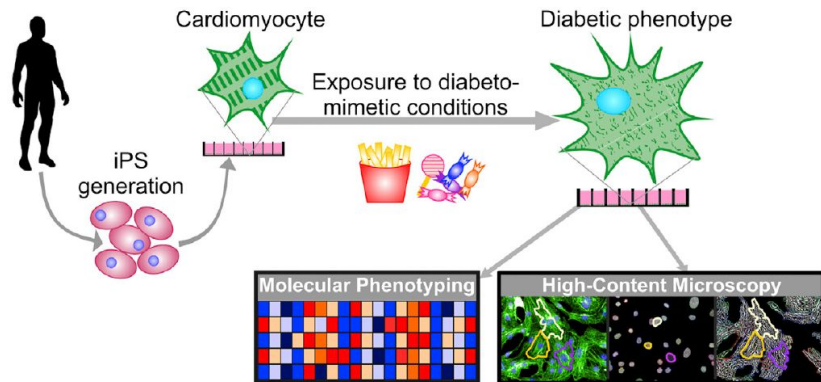
Gene-set enrichment analysis



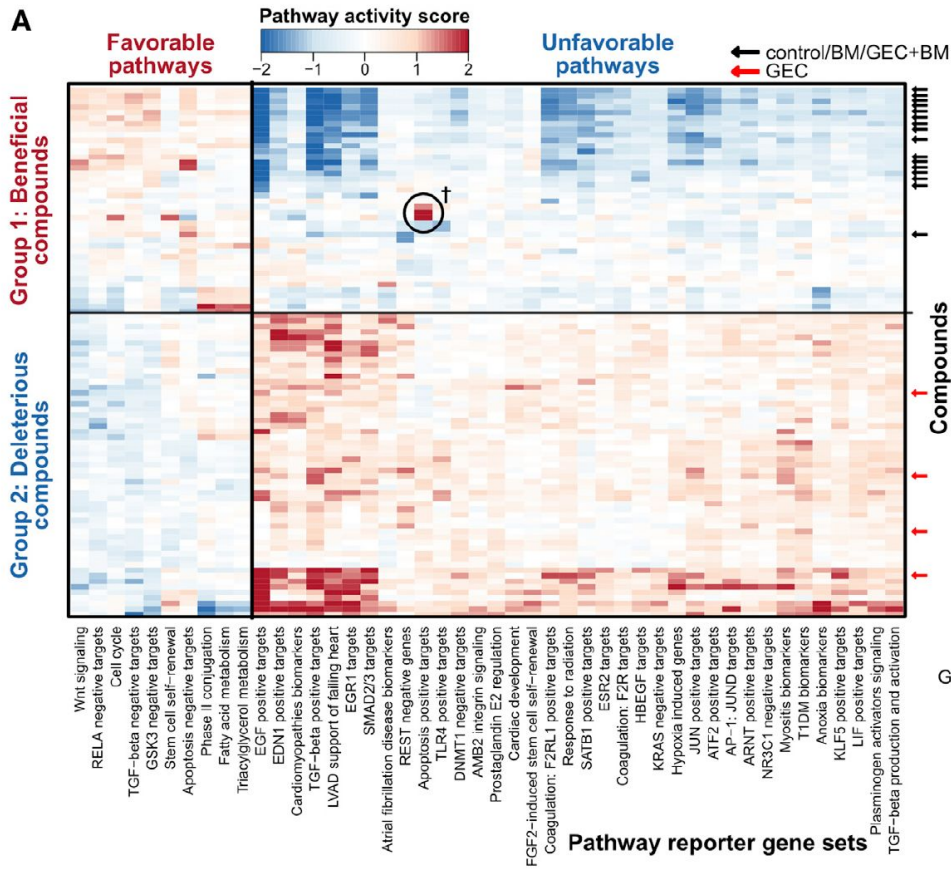
Input: (1) a differential gene expression profile; (2) a set of gene-sets $\{G\}$, each a set of genes.

Output: a ranked list of the input gene-sets by *enrichment*.

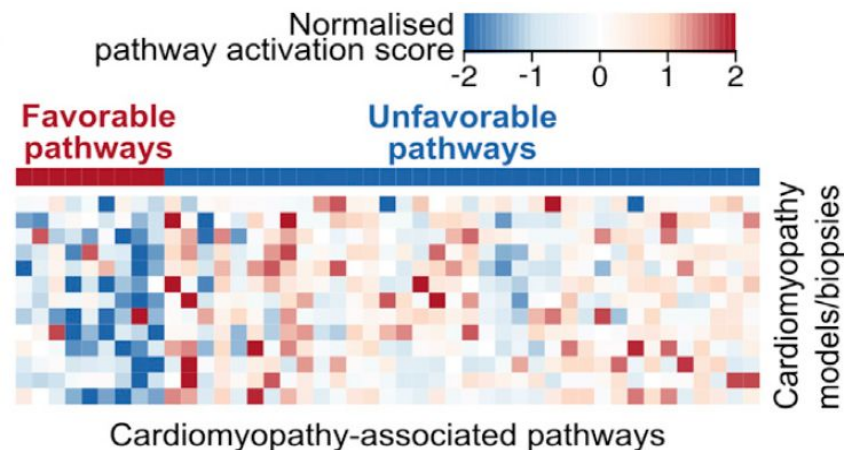
Gene expression as screening readout



Differential gene expression profiles are molecular snapshots of drugs' action in the cell.



Gene expression from patient and animal models help compound selection



We can prioritise molecules that reverse disease-induced changes.

